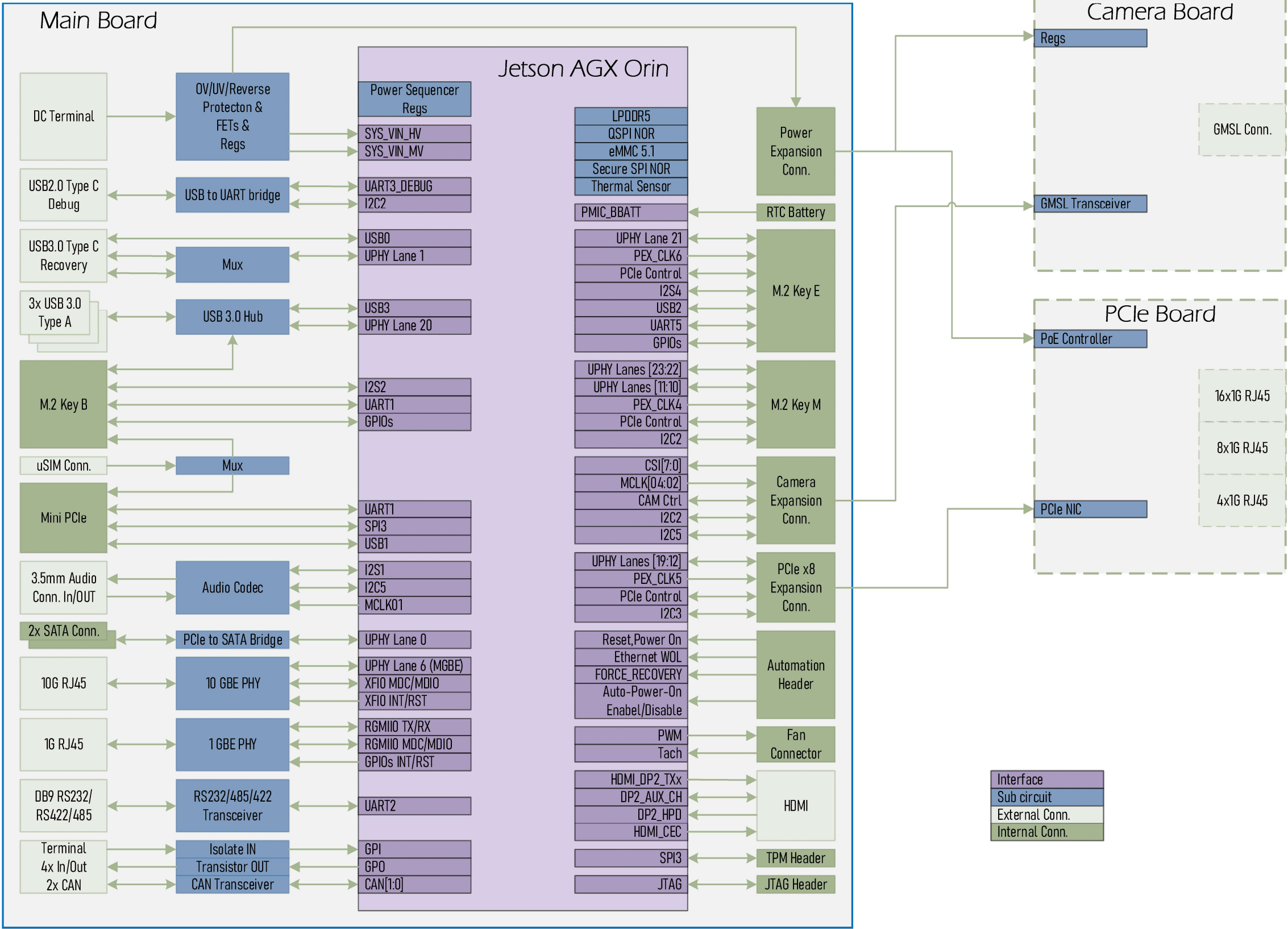


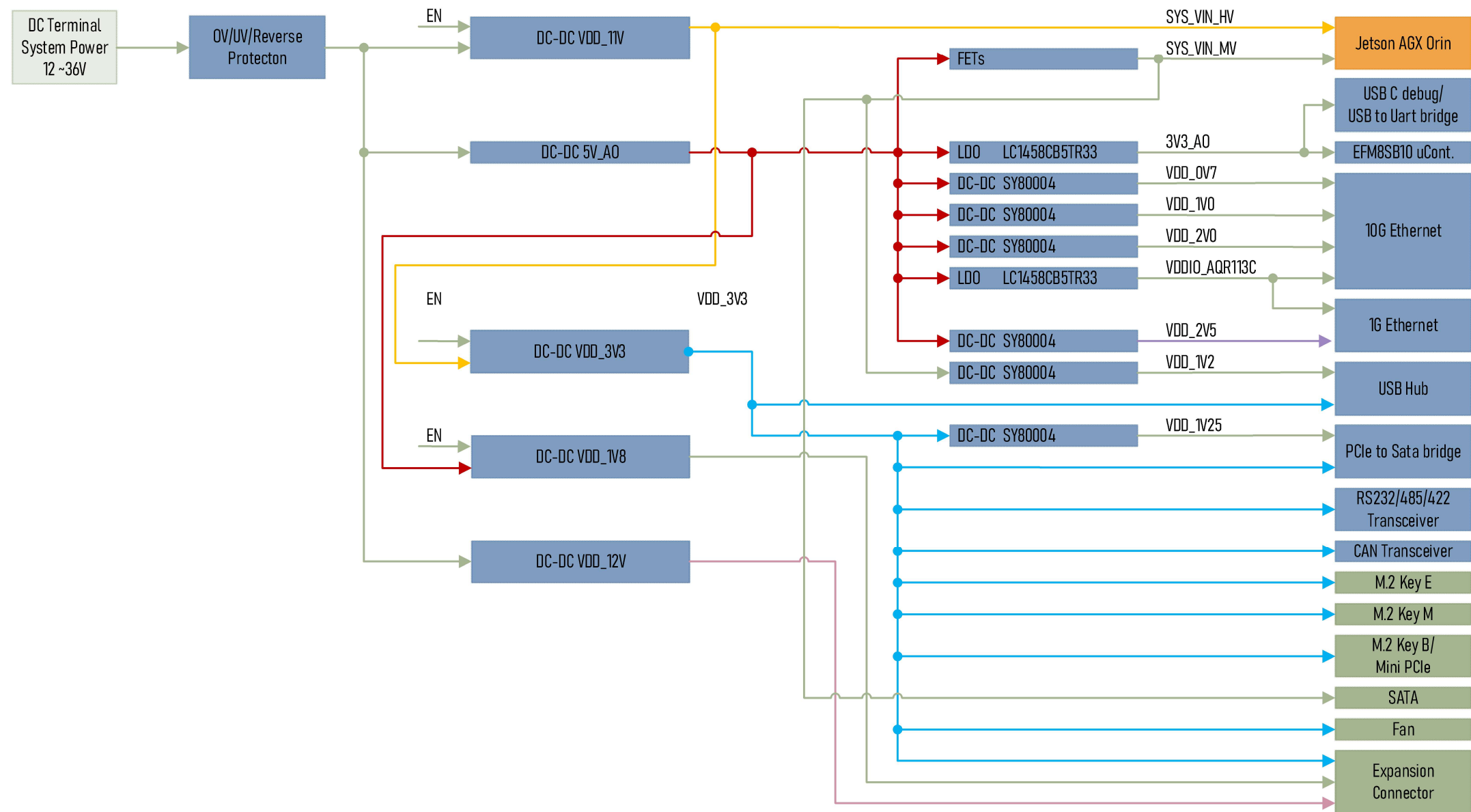
Schematic: reServer Industrial J501

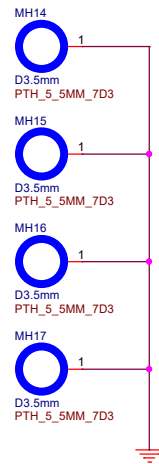
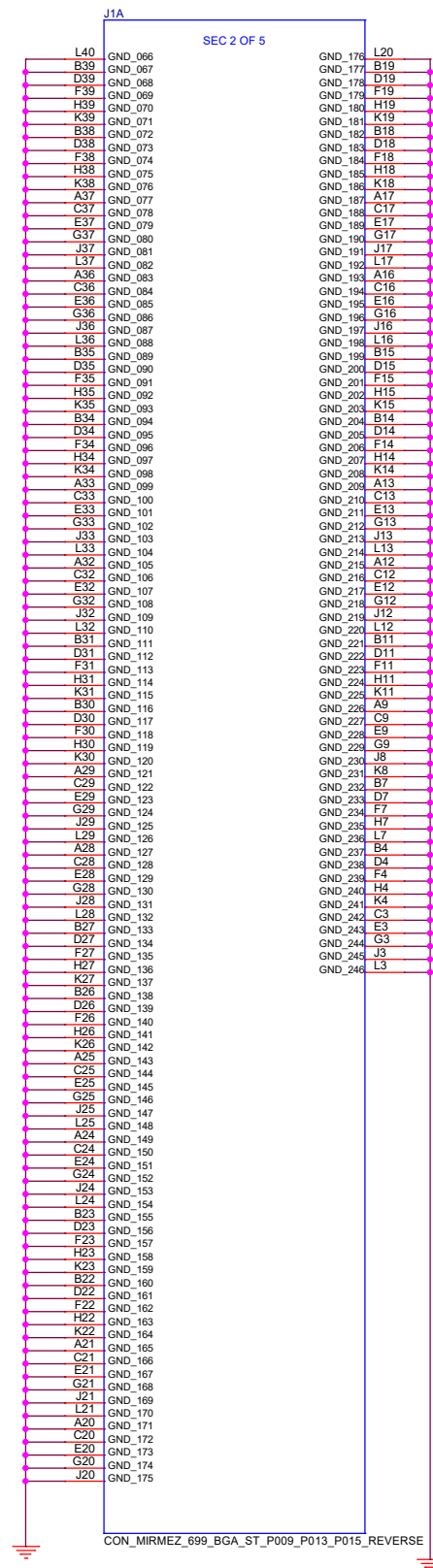
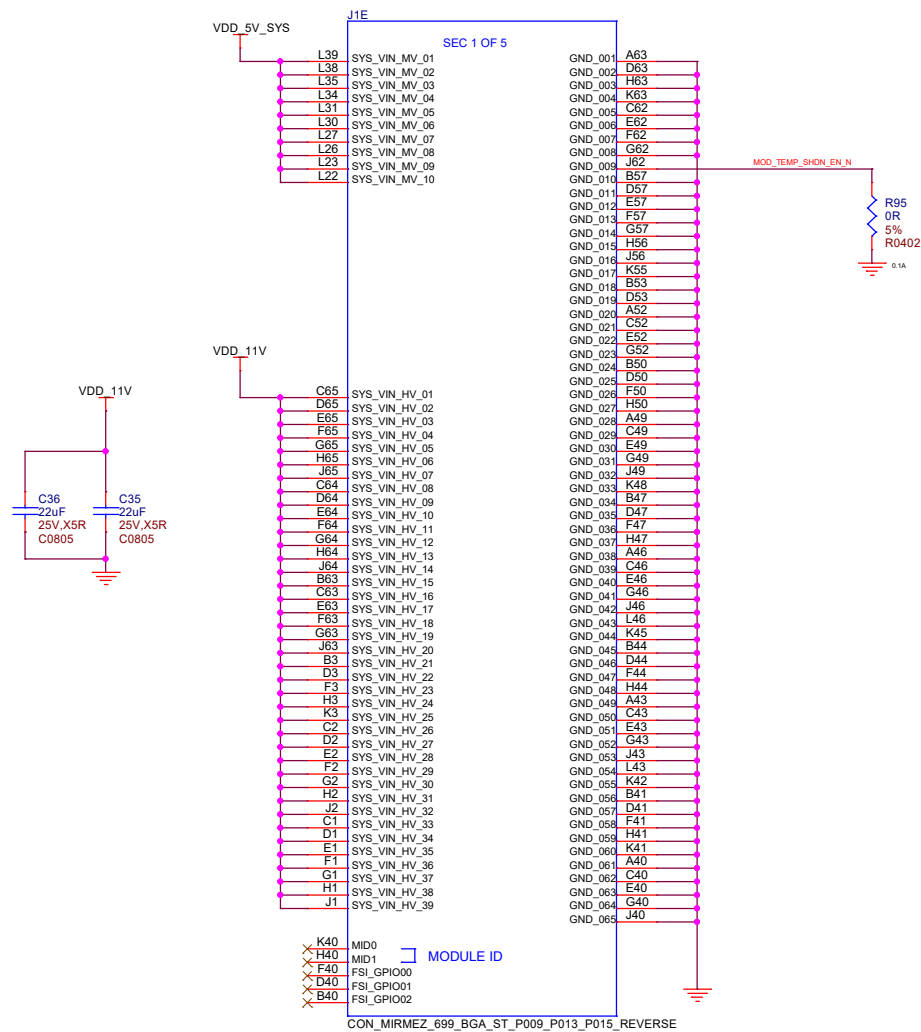
SHEET	SHEET NAME
01	01 Table of Contents
02	02 Block Diagram
03	03 Power Tree
04	04 CVM Connector 1/3
05	05 CVM Connector 2/3
06	06 CVM Connector 3/3
07	07 Power Vin 5V 3V3_A0 1V8 0V7
08	08 Power 12V 11V 3V3 2V 1V2 1V
09	09 Button MCU For Power Up
10	10 10G Ethernet
11	11 Gigabit Ethernet
12	12 USB3.1 HUB
13	13 USB3.1 Type-A x2 (A)
14	14 USB3.1 Type-A x1(B)
15	15 Type C, Debug UART
16	16 M.2 KEY-E (WIFI/BT)
17	17 M.2 KEY-M (NVME)
18	18 M.2 KEY-B (4G/5G)
19	19 Mini PCIe (4G/LoRa)
20	20 HDMI
21	21 Fan, EEPROM, Debug
22	22 DI, DO
23	23 CAN, Isolated 5V
24	24 TPM, IIC to IO
25	25 RS232/422/485
26	26 SATA_Controller
27	27 SATA_Connector
28	28 Audio Codec
29	29 Camera Expansion
30	30 PCIe Expansion

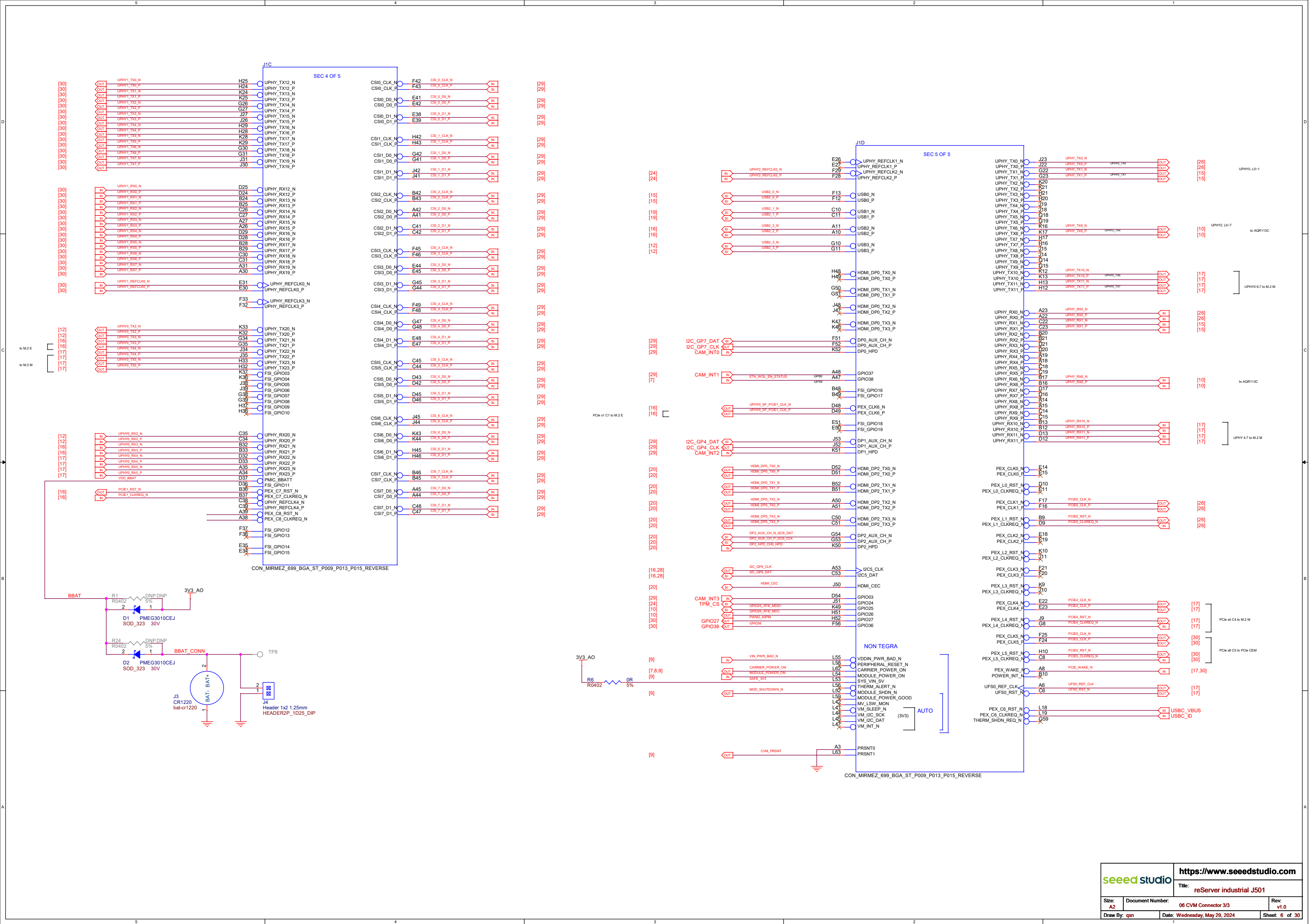
Revision History

DATE	REVISION	DESCRIPTION
May 15, 2024	v1.0	First release.

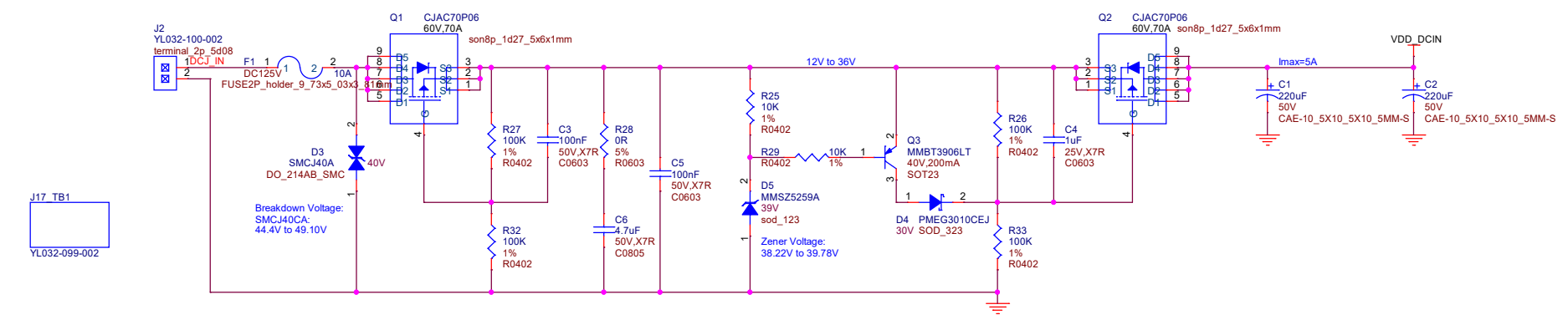




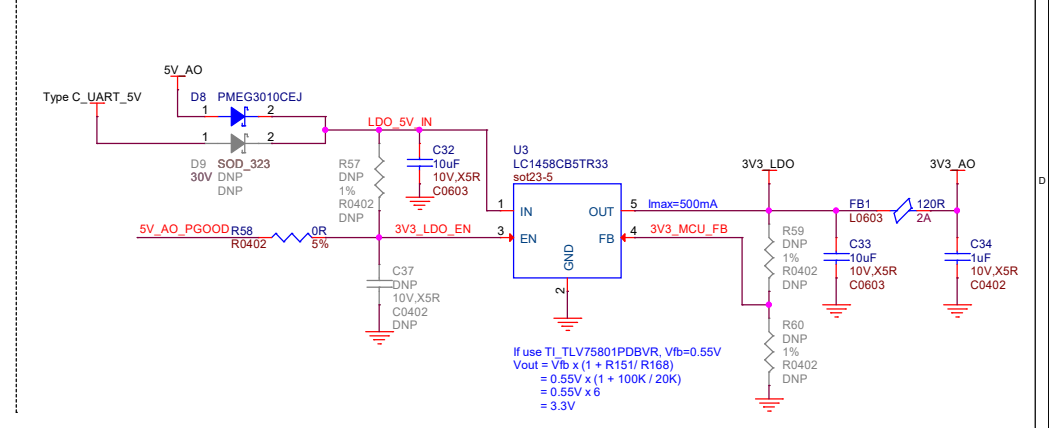




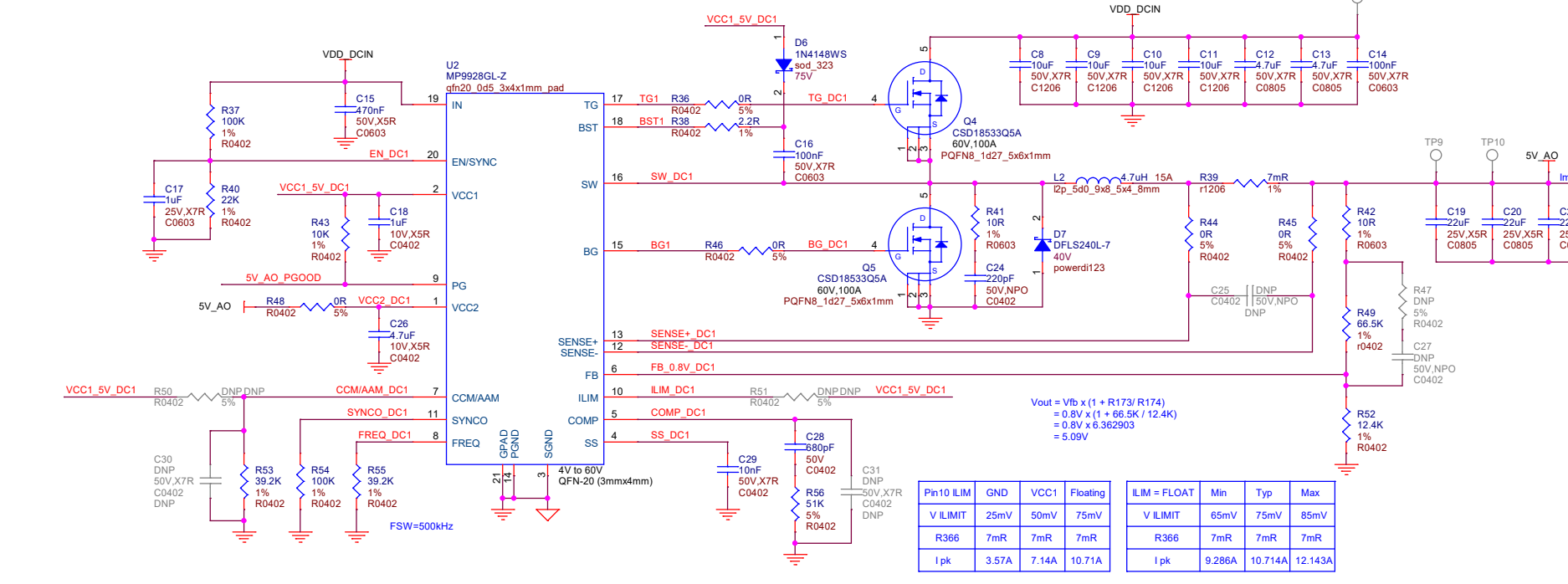
Vin protection



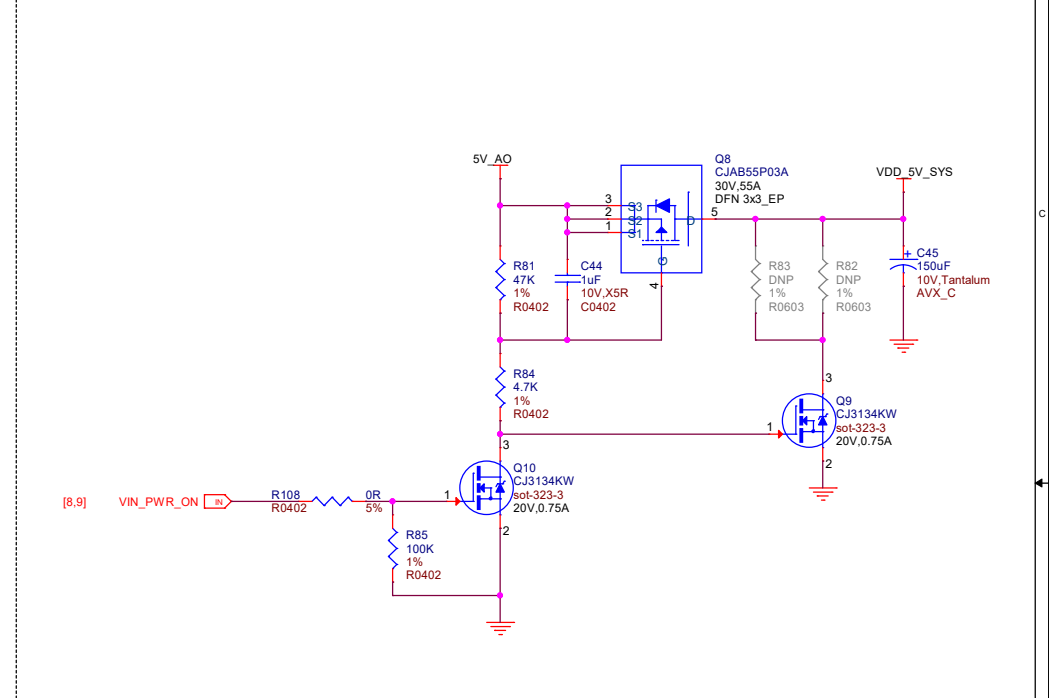
3V3_AO



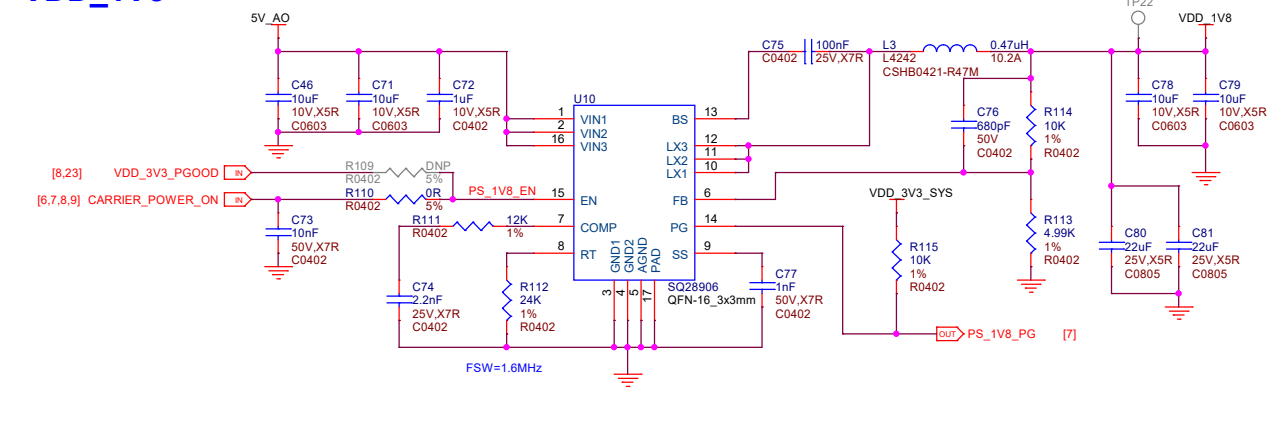
5V_AO



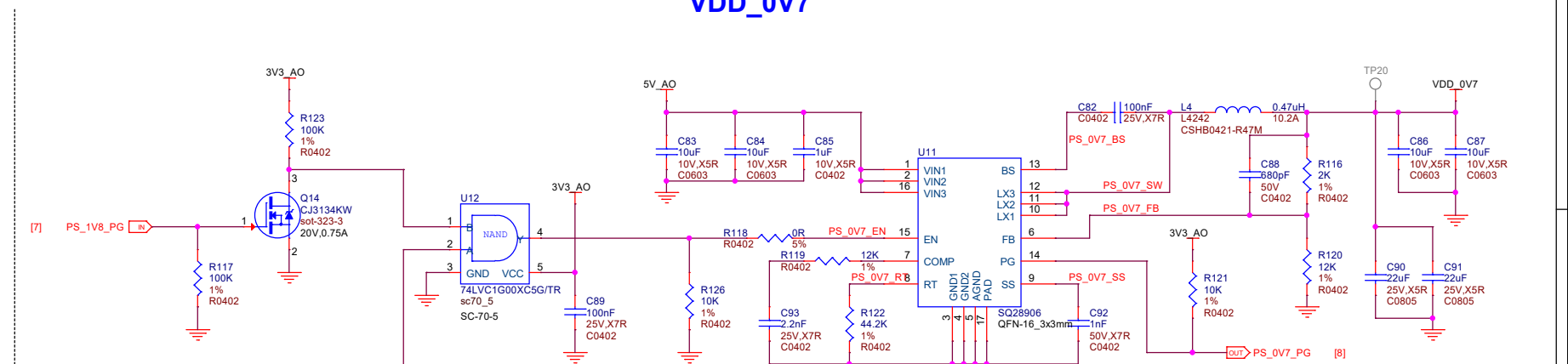
VDD_5V



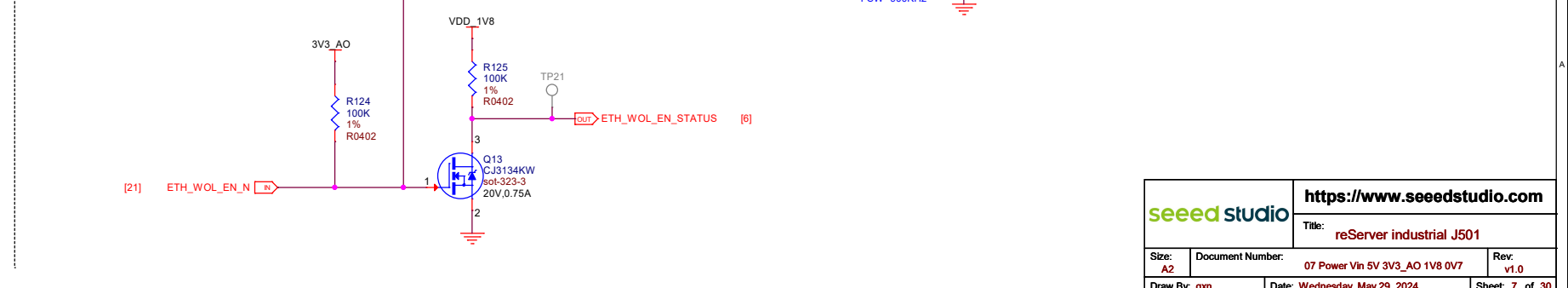
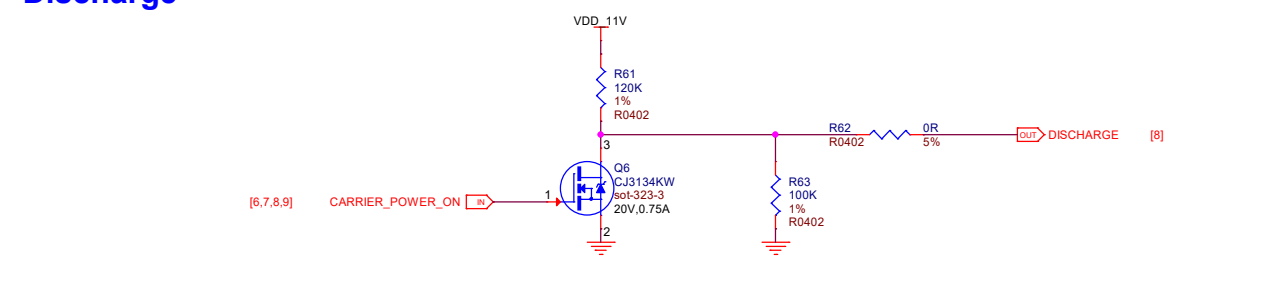
VDD_1V8



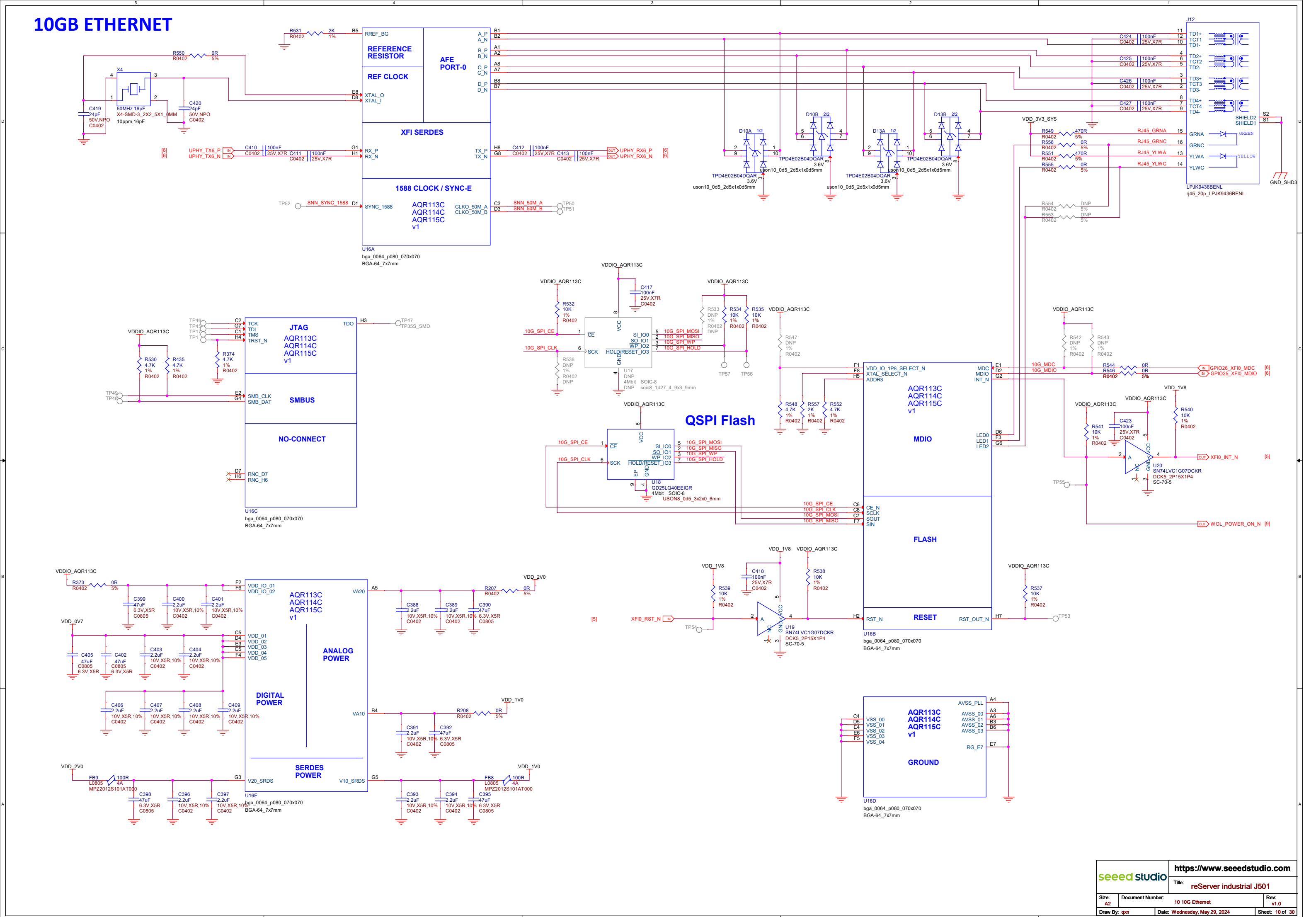
VDD_0V7



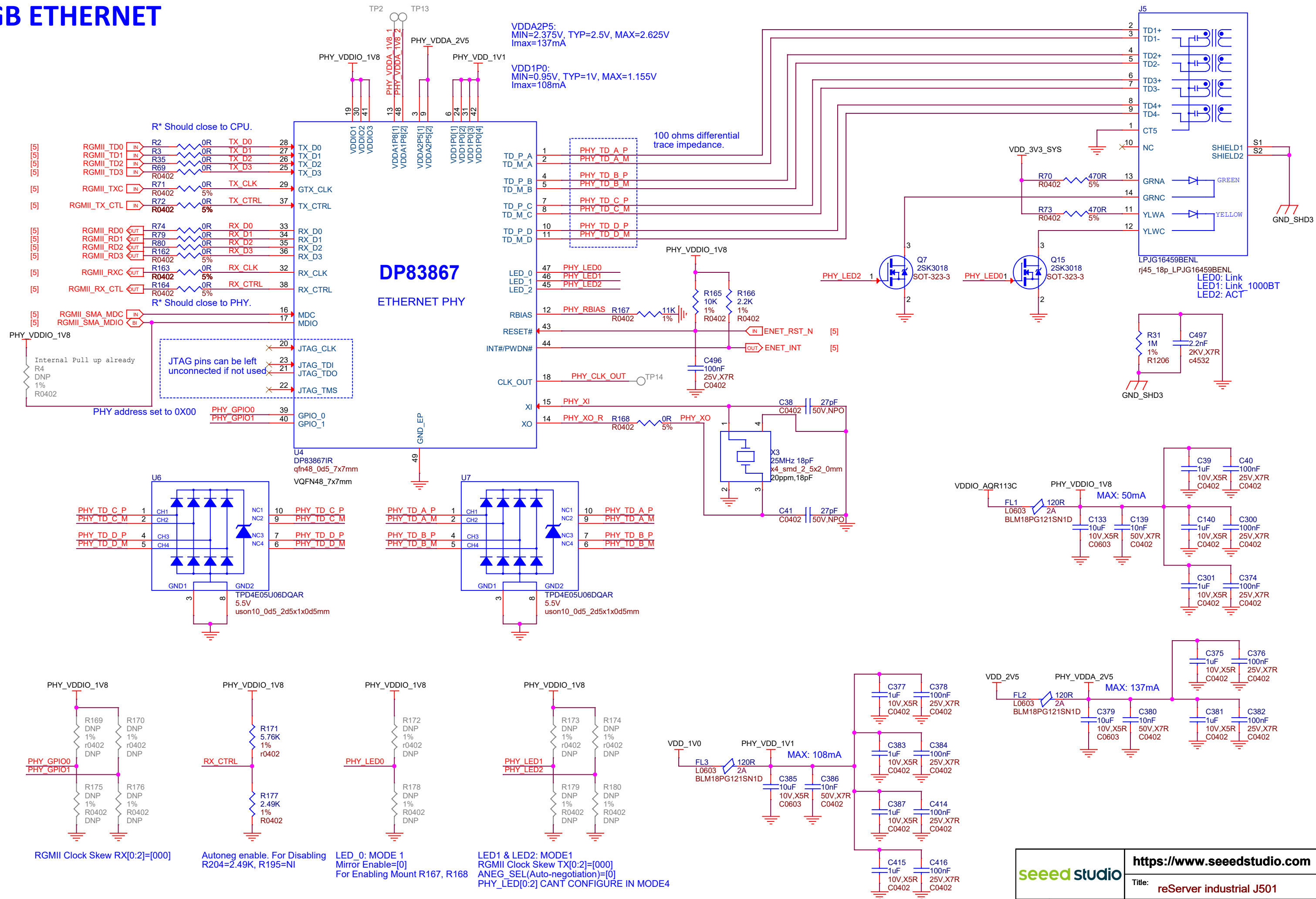
Discharge



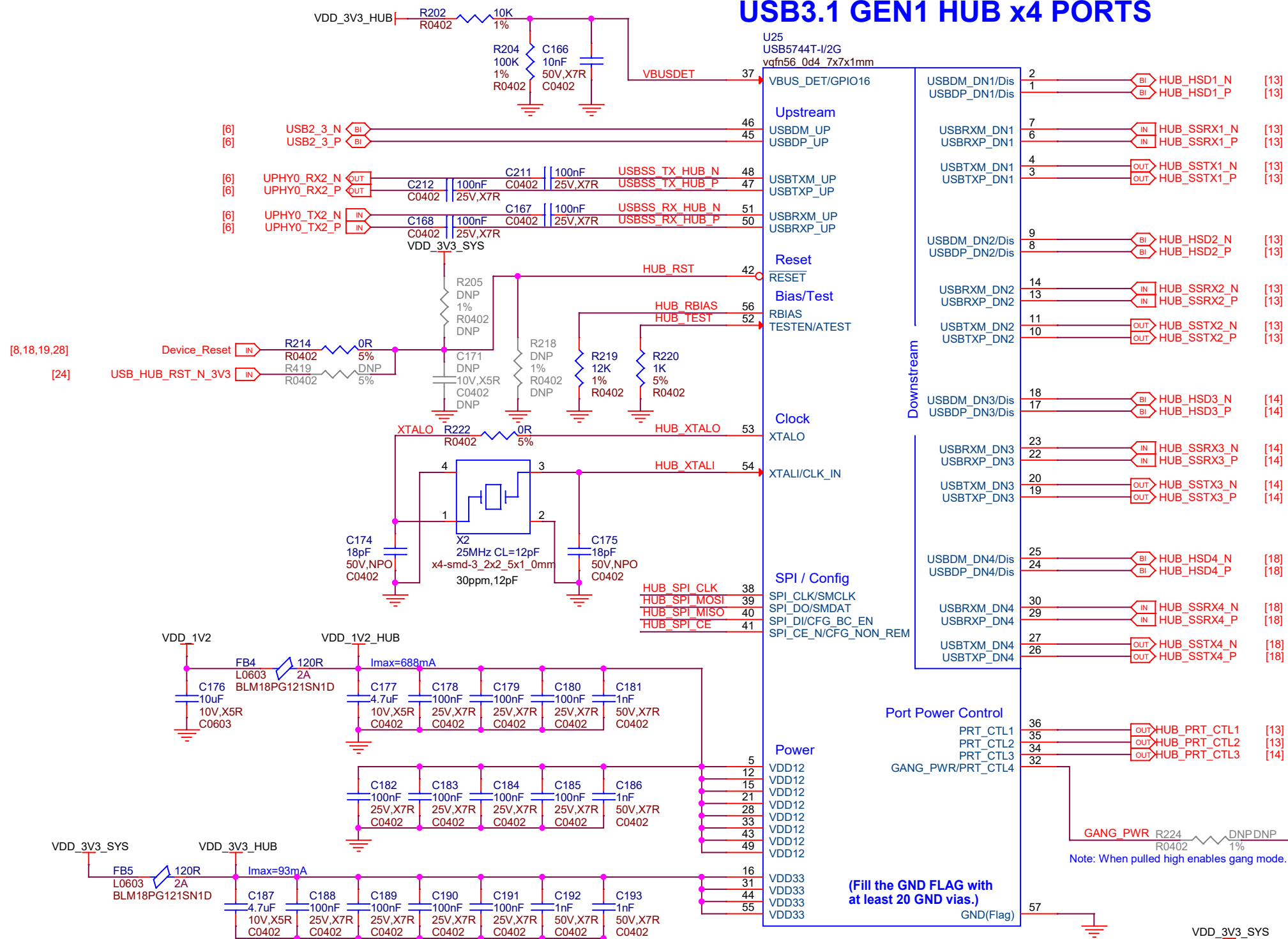
10GB ETHERNET



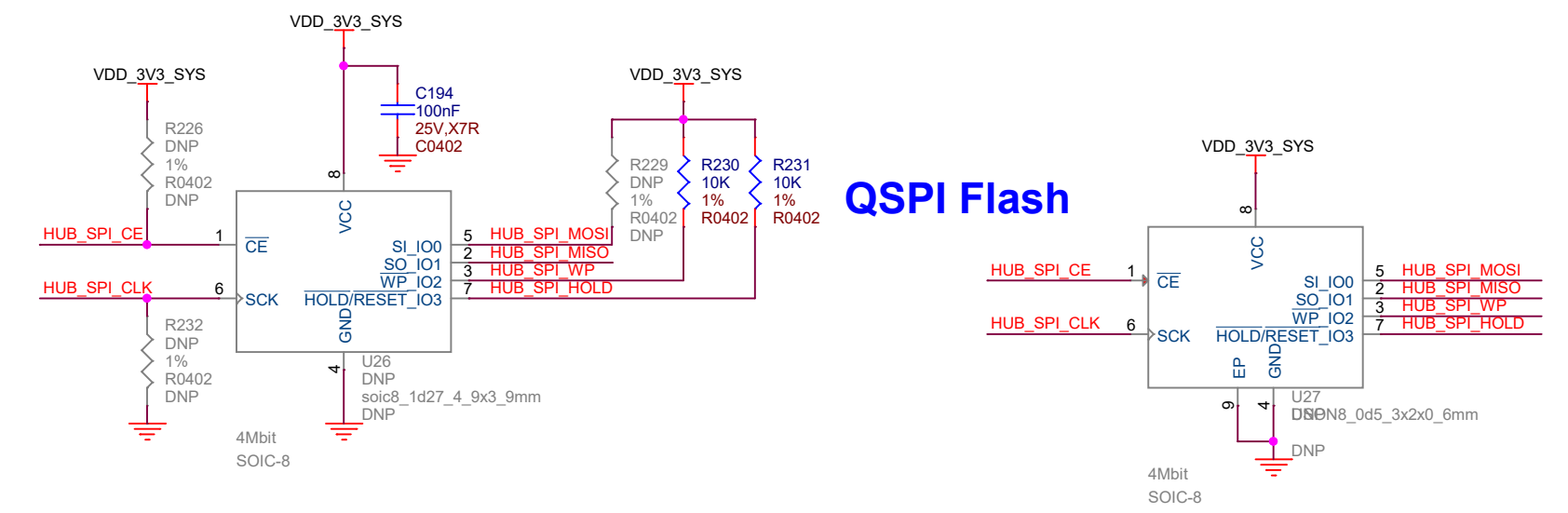
GB ETHERNET

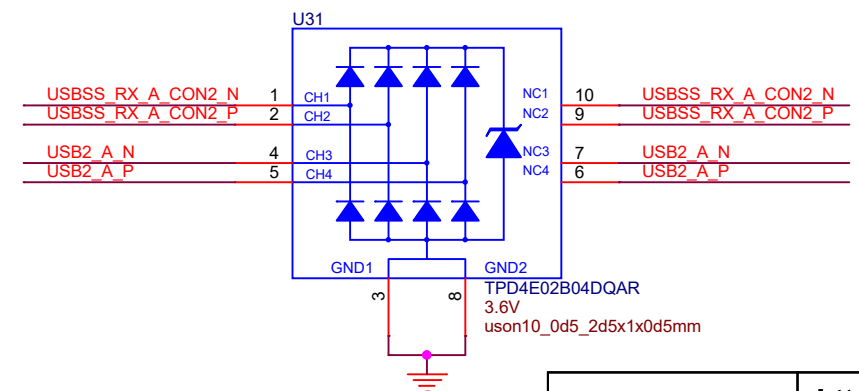
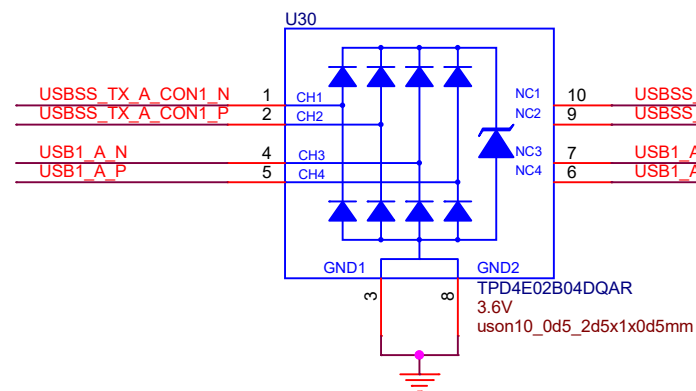
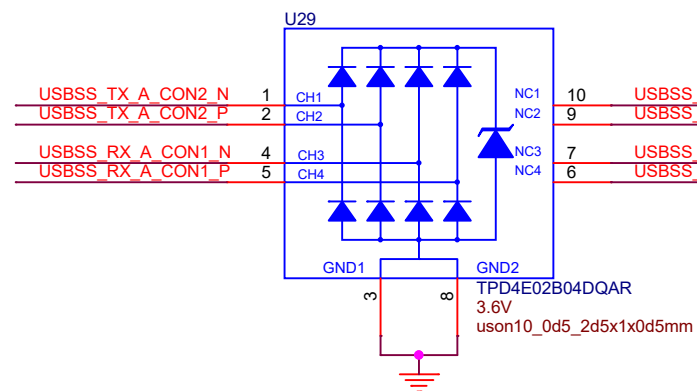
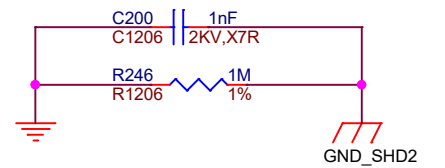
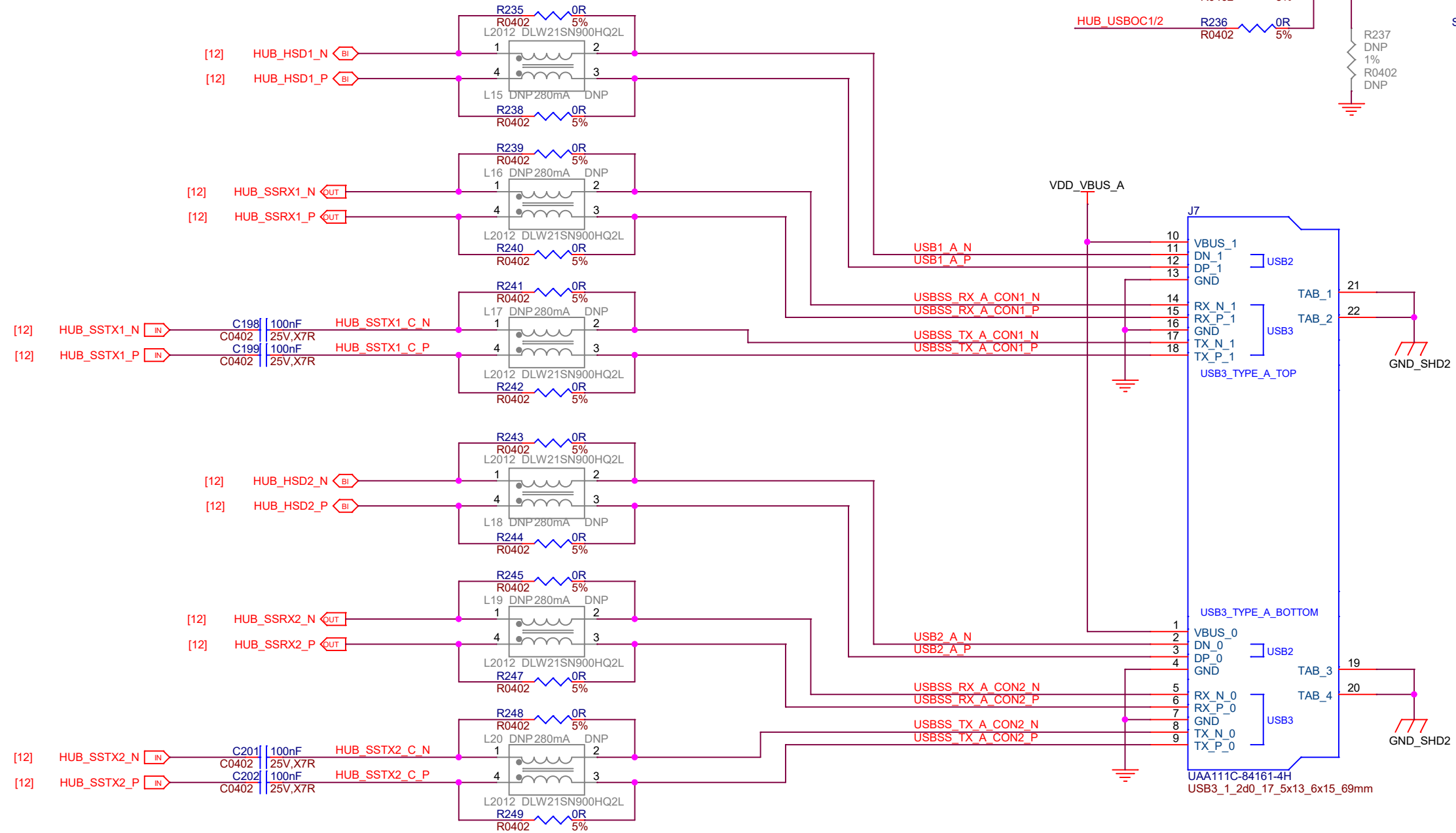


USB3.1 GEN1 HUB x4 PORTS

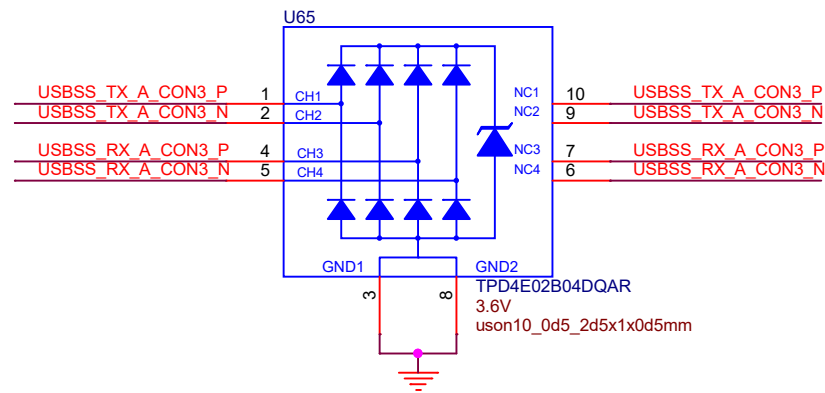
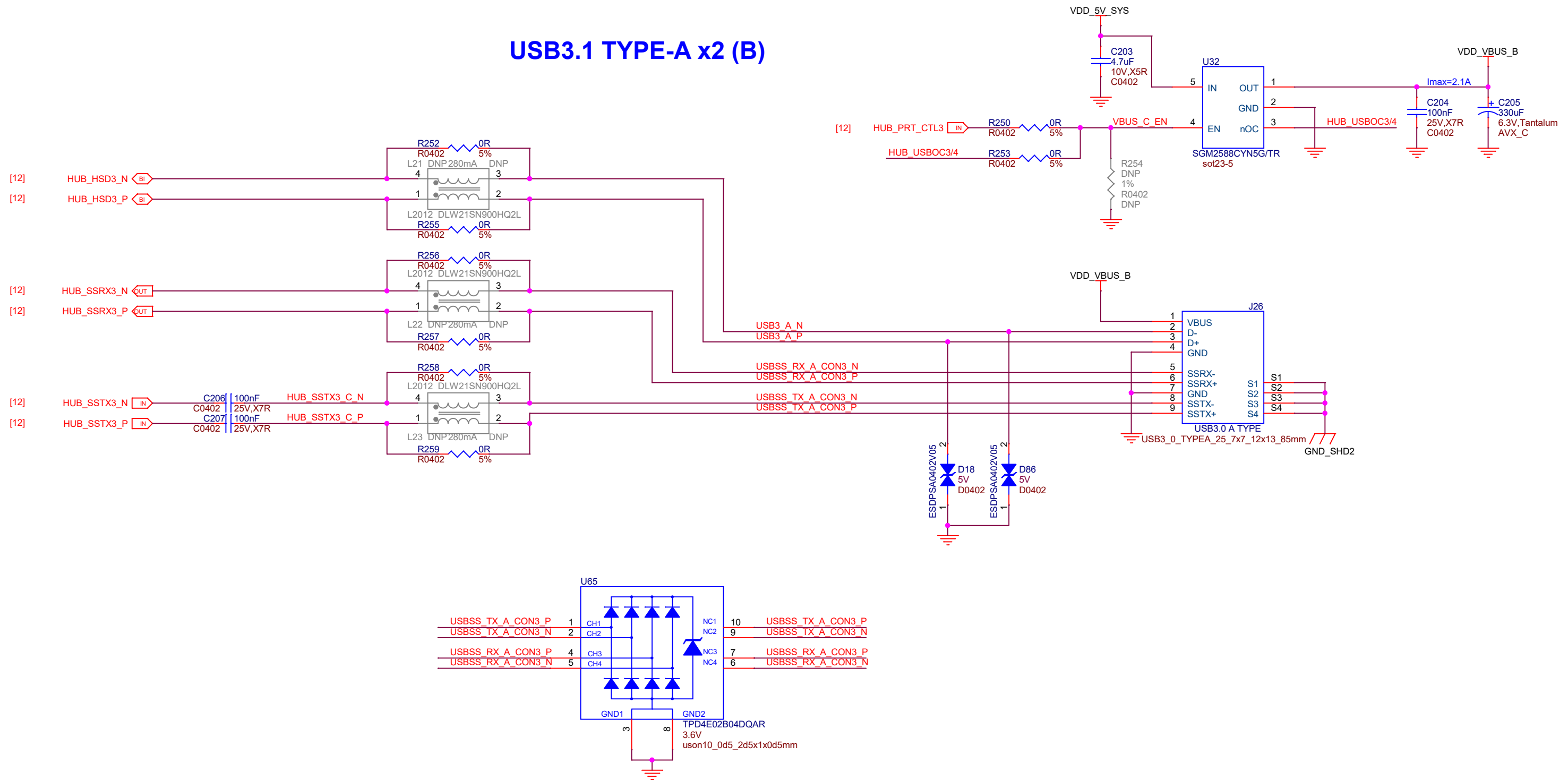


QSPI Flash



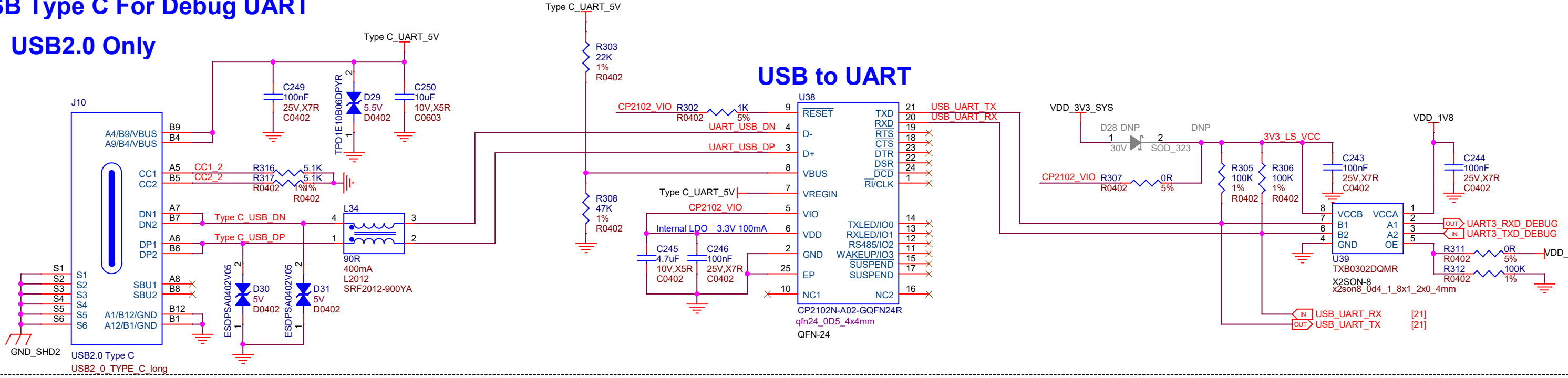


USB3.1 TYPE-A x2 (B)

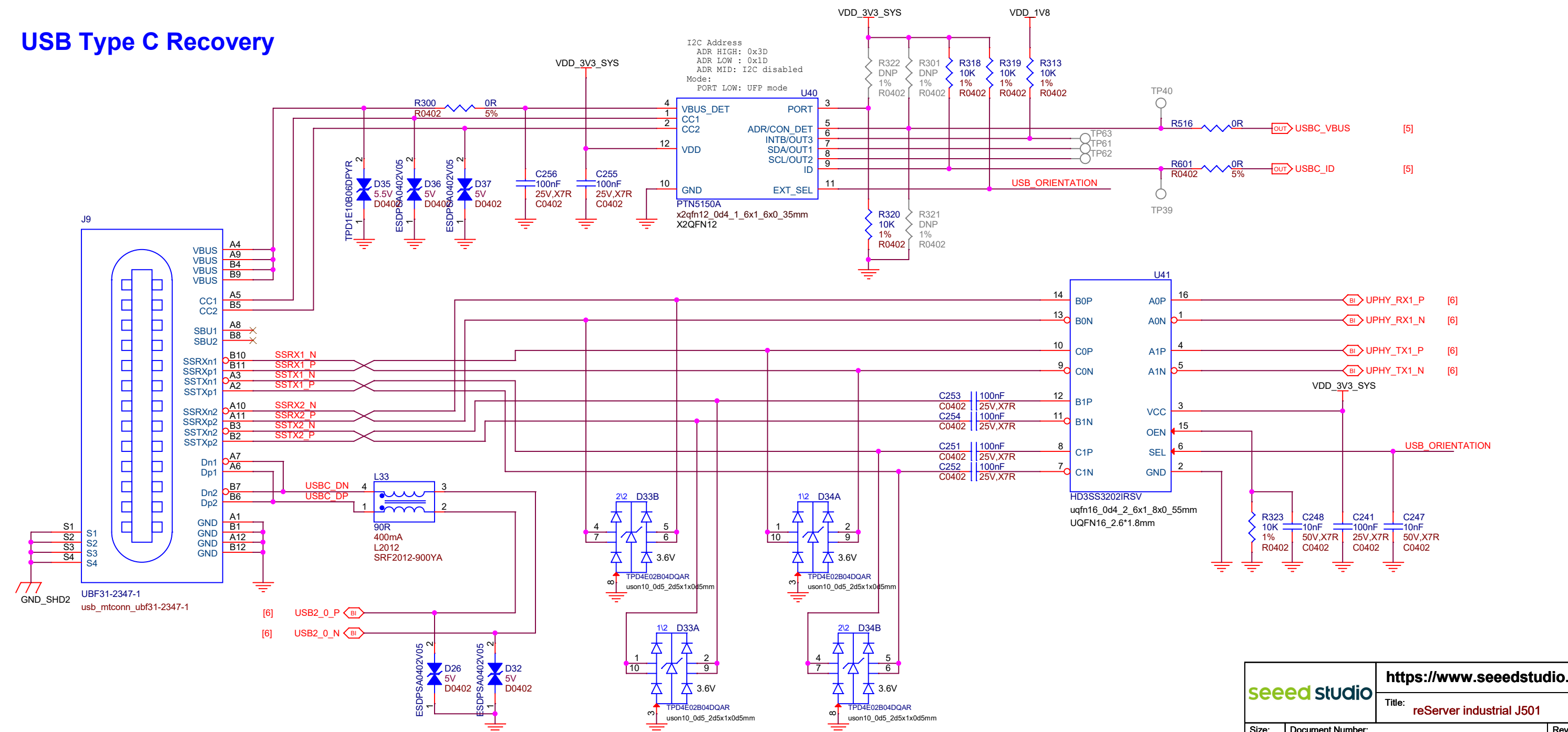


USB Type C For Debug UART

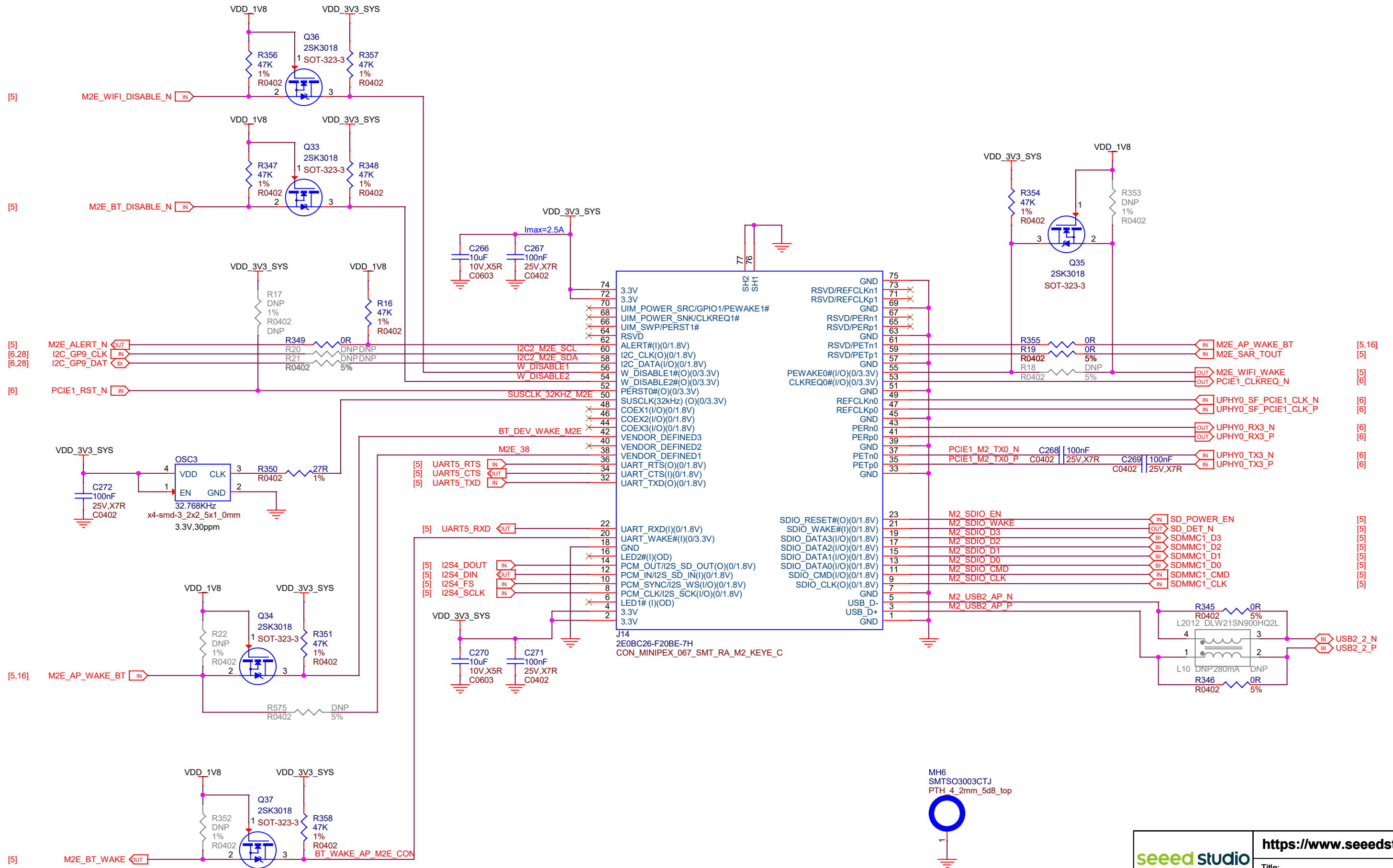
USB2.0 Only



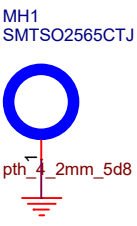
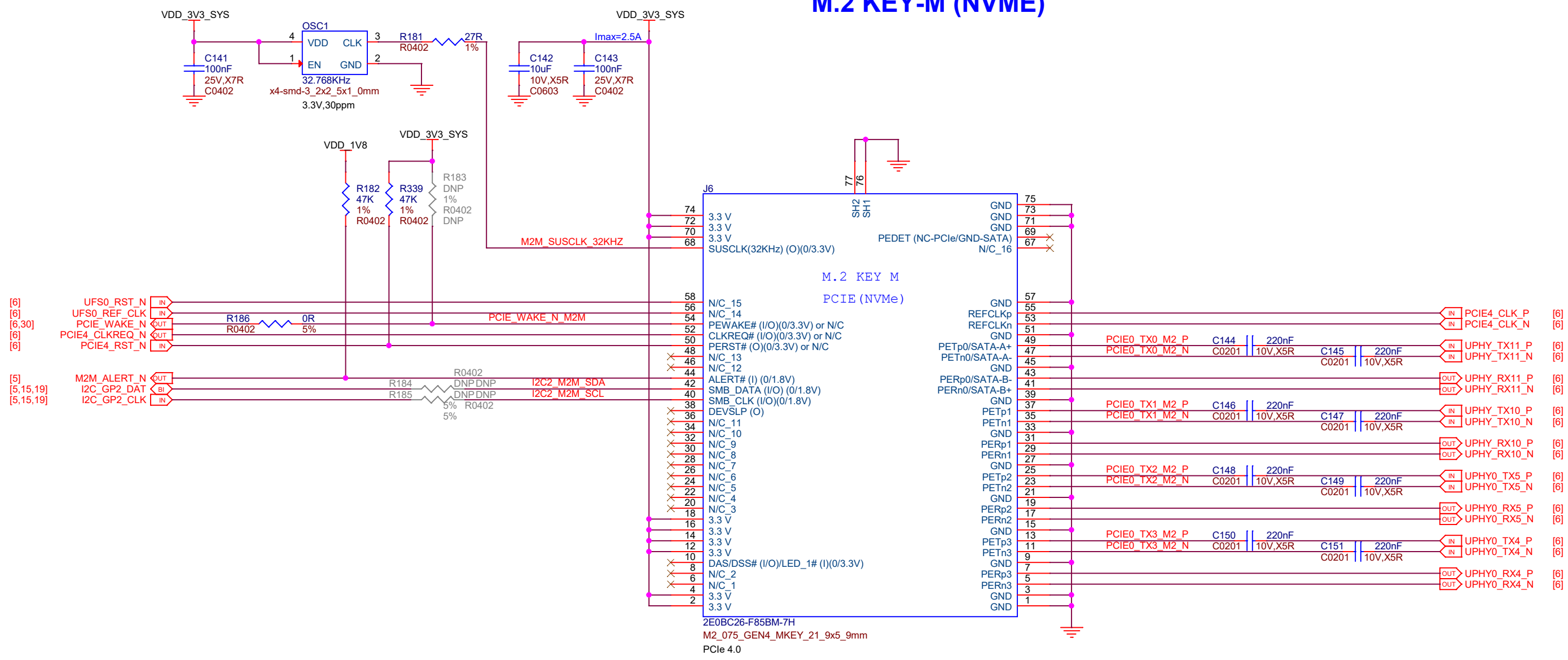
USB Type C Recovery

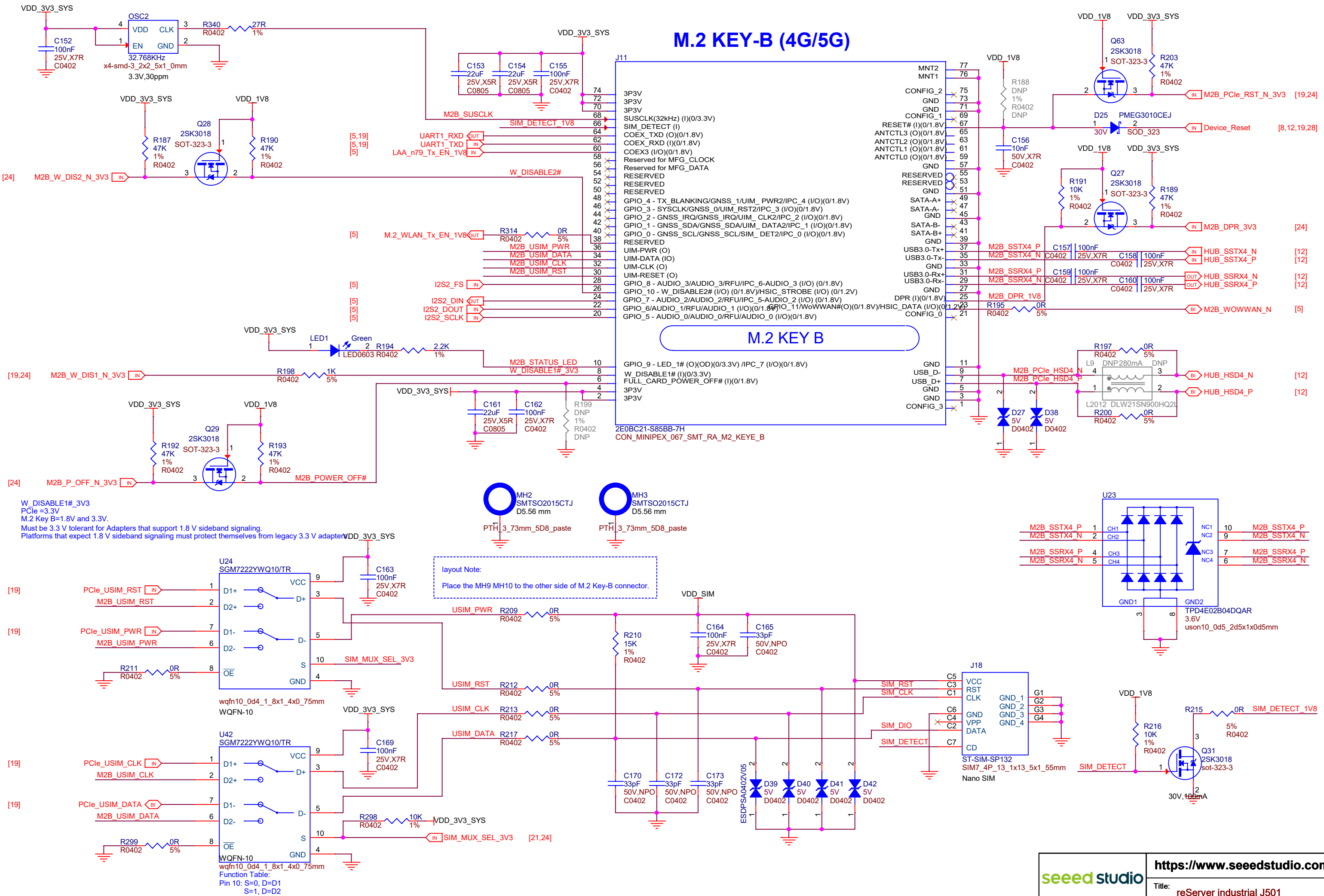


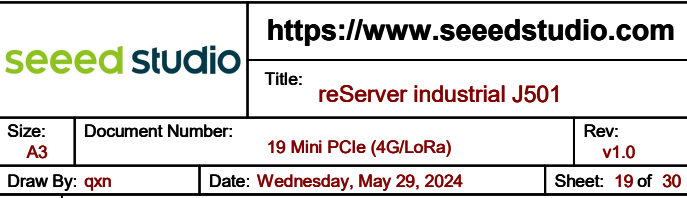
M.2 KEY-E (WIFI/BT)



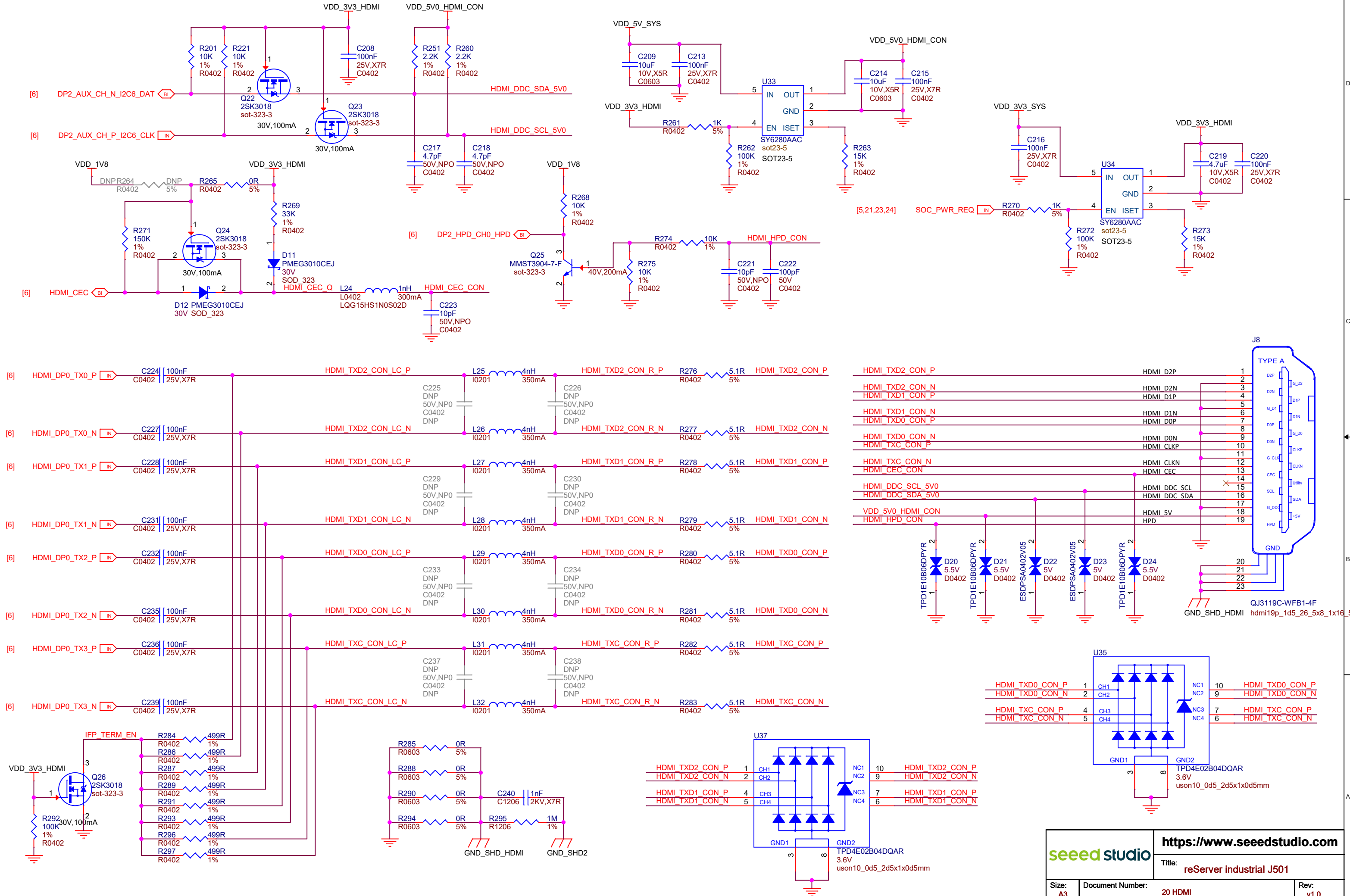
M.2 KEY-M (NVME)



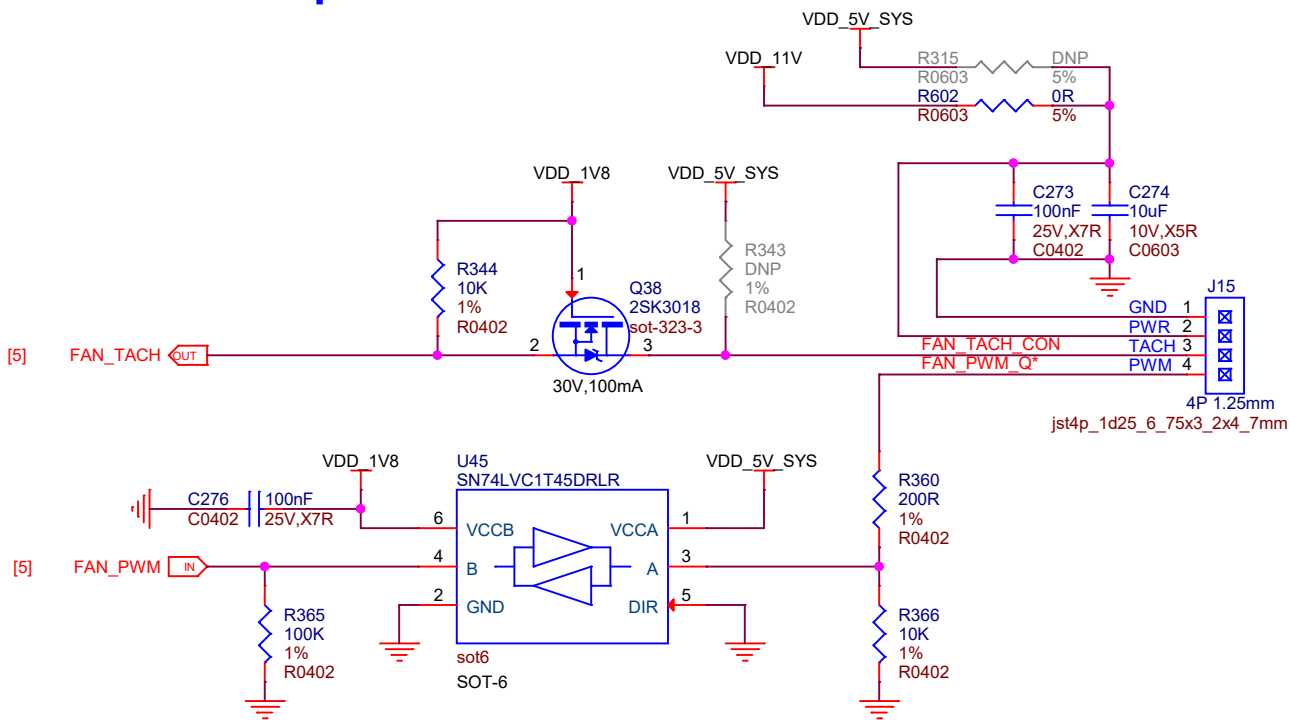




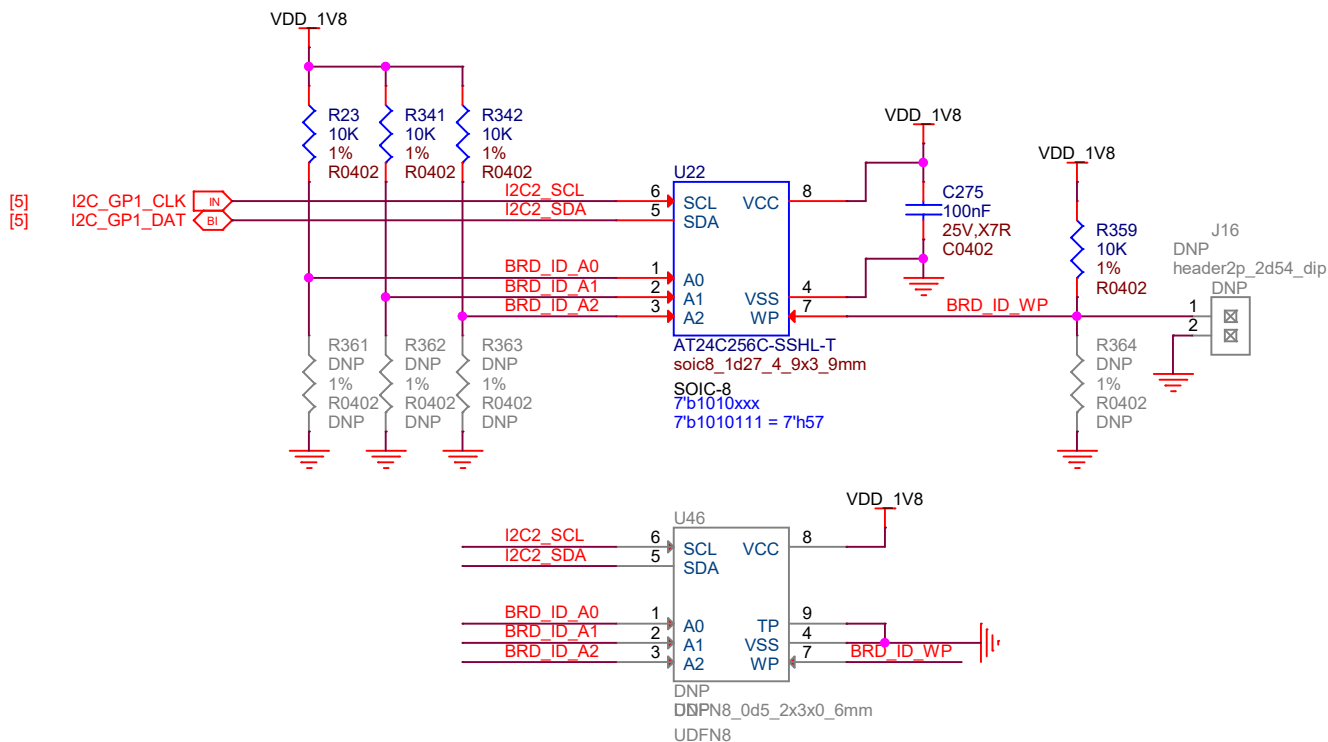
HDMI Connector



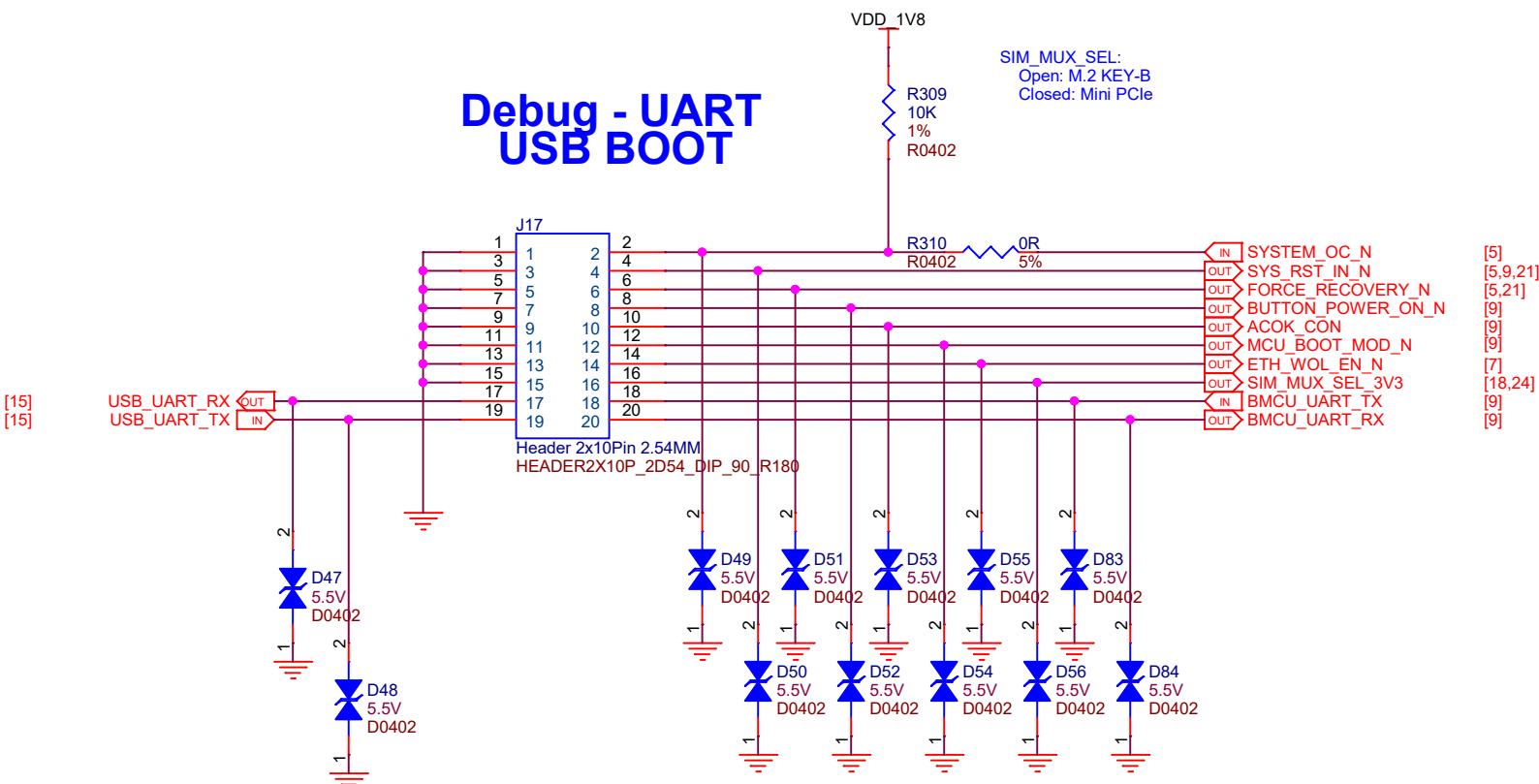
SoC Fan Header 4pin



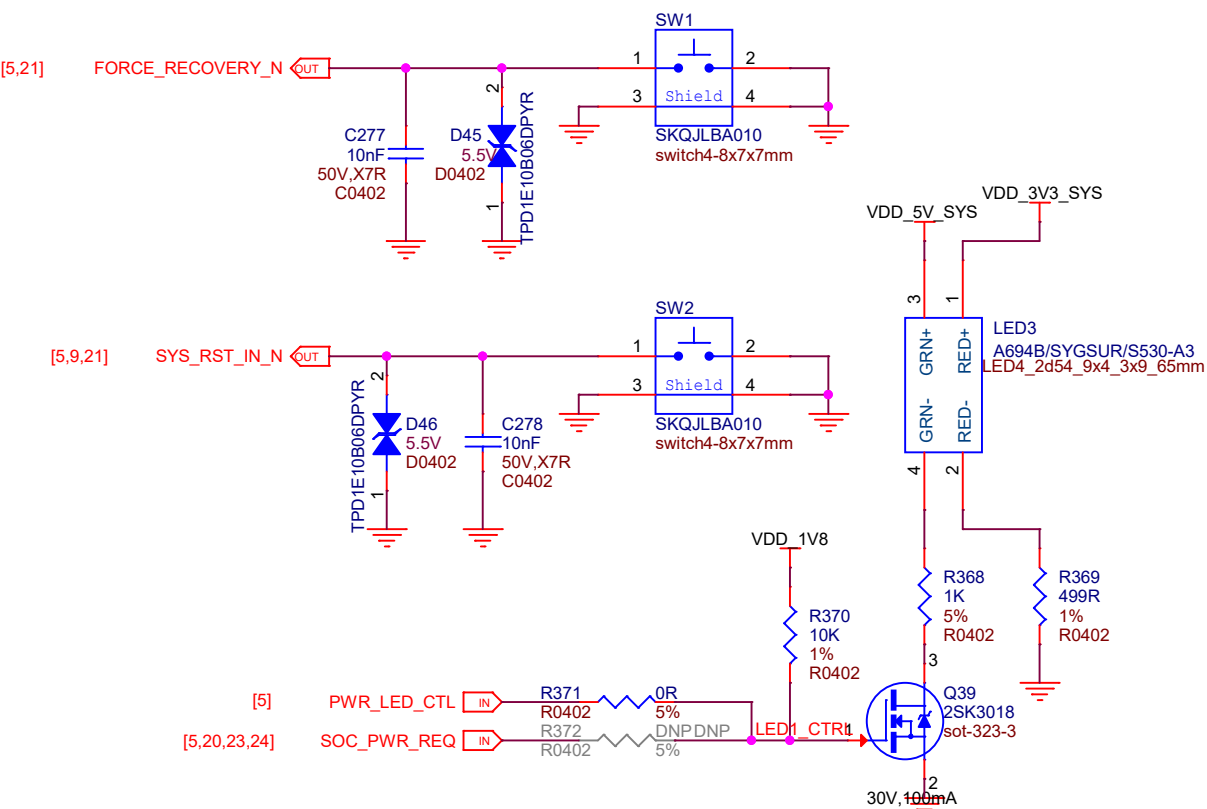
Carrier Board Config



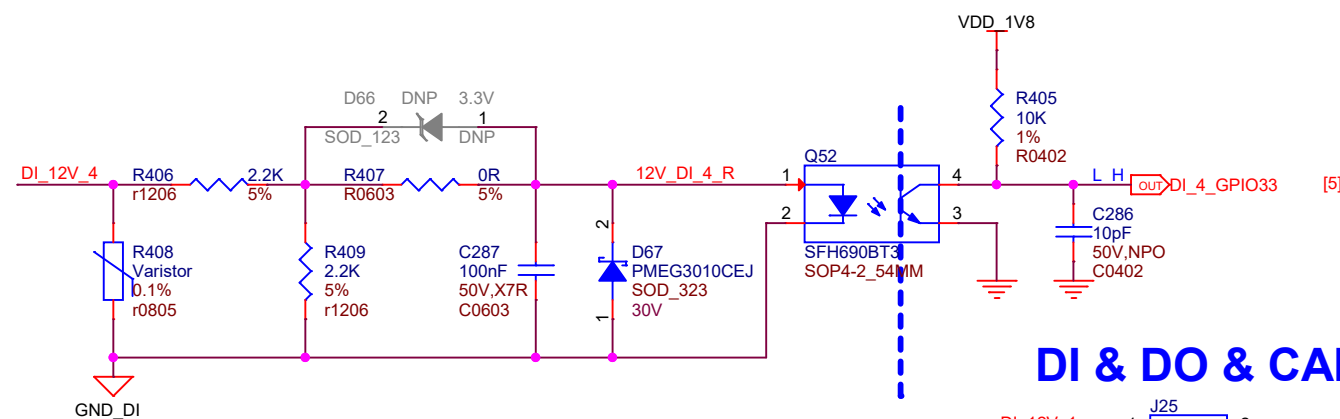
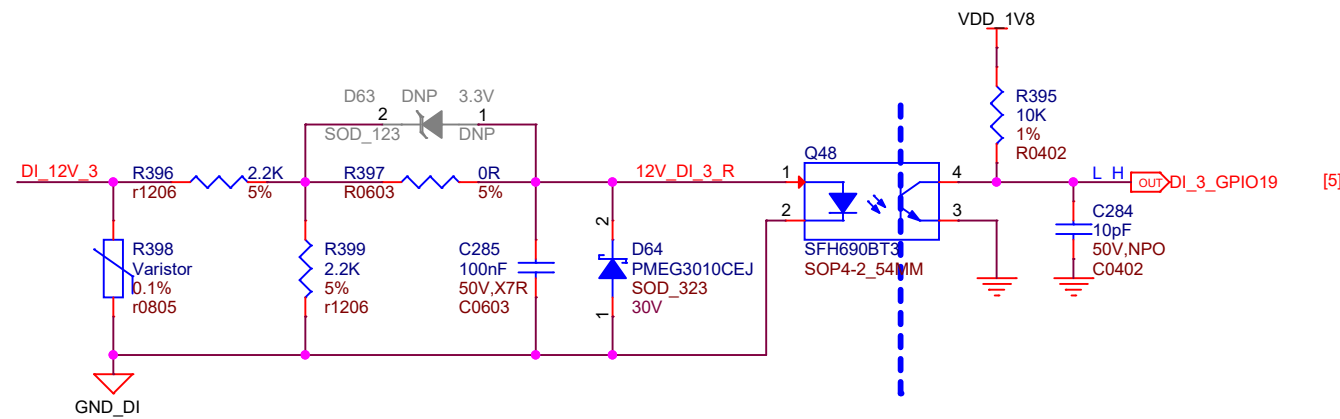
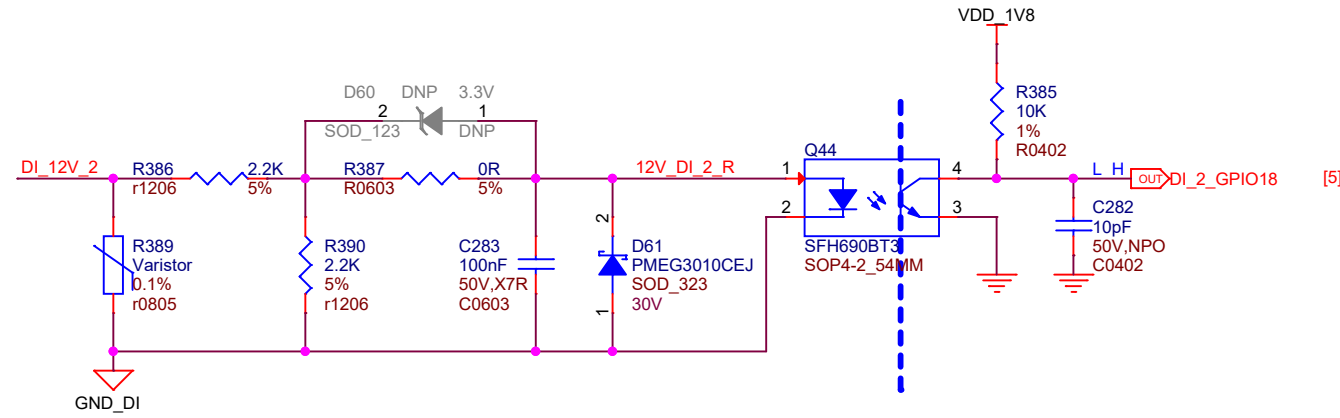
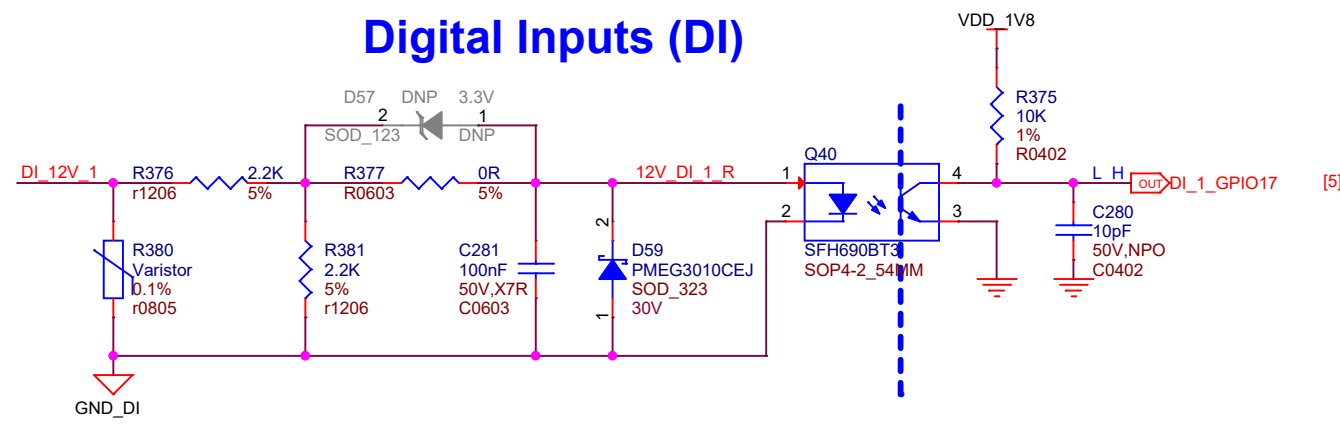
Debug - UART
USB BOOT



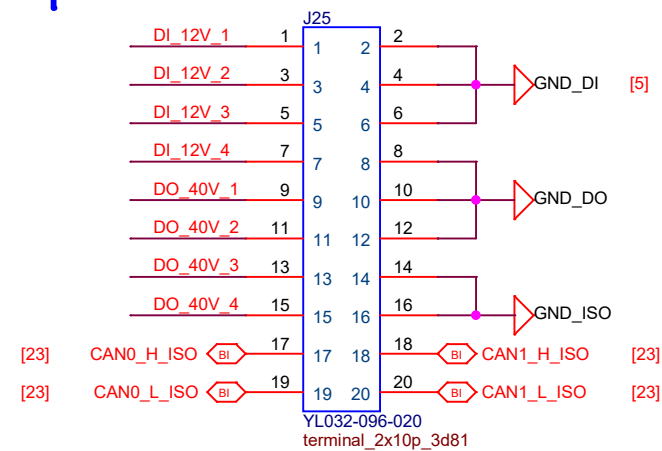
LED & Button



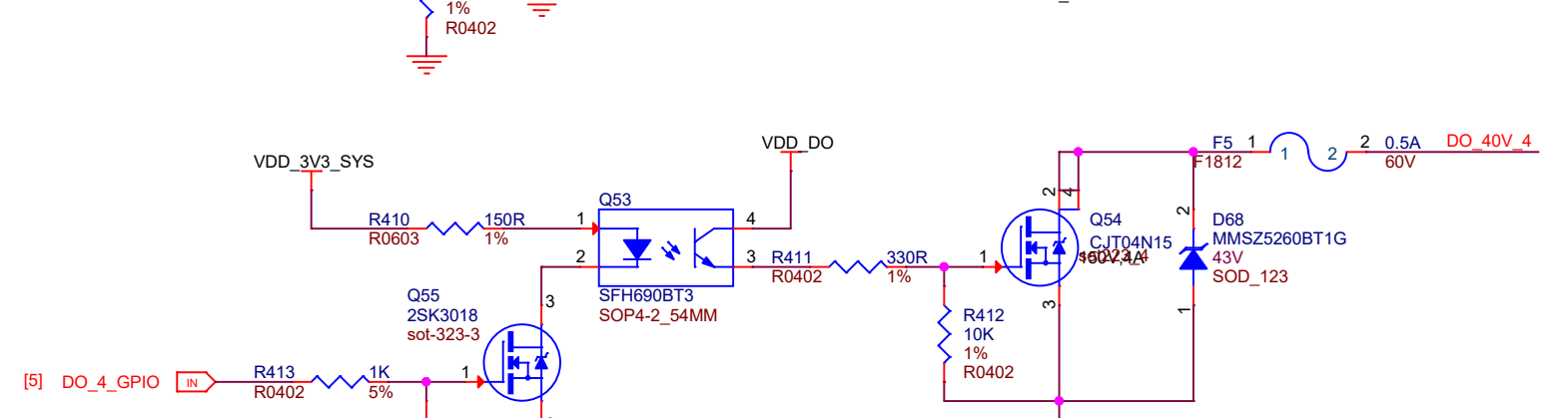
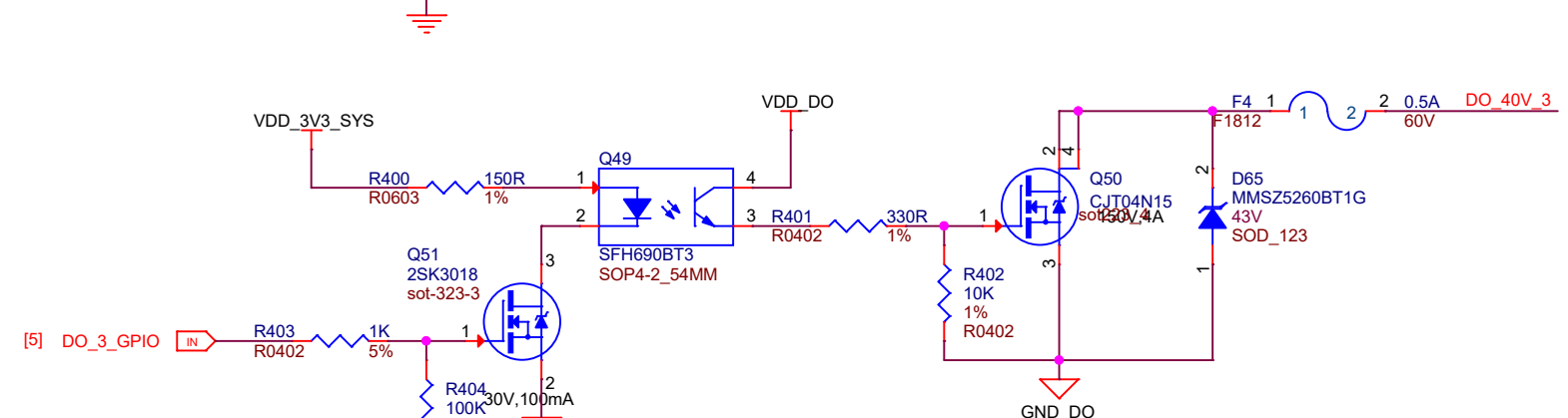
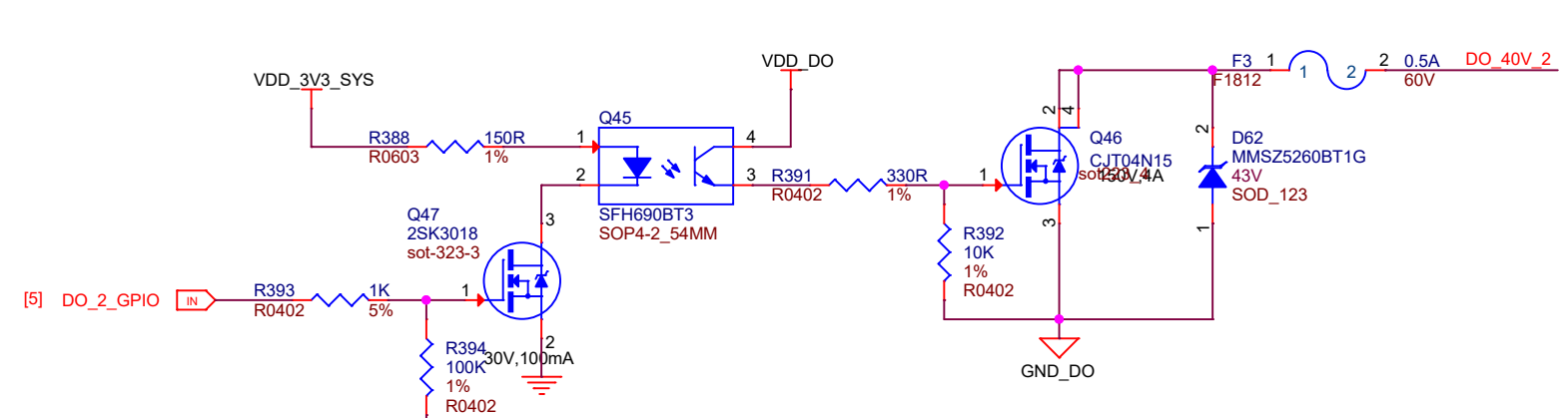
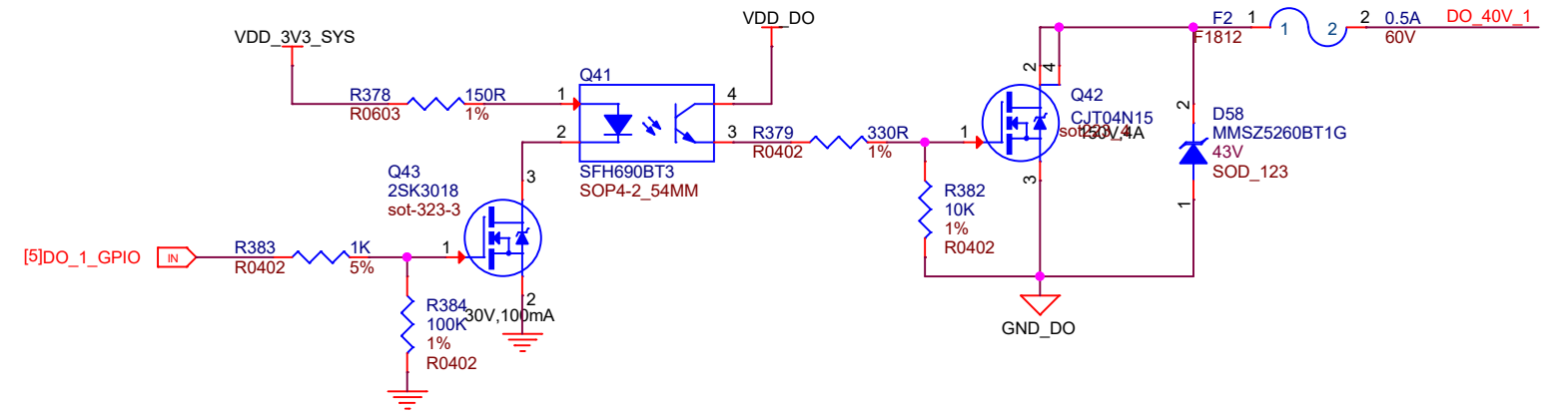
Digital Inputs (DI)



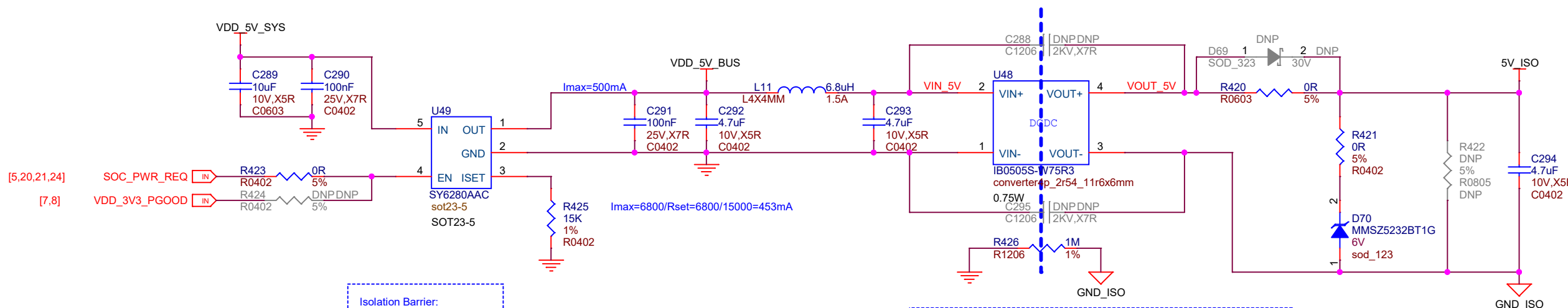
DI & DO & CAN



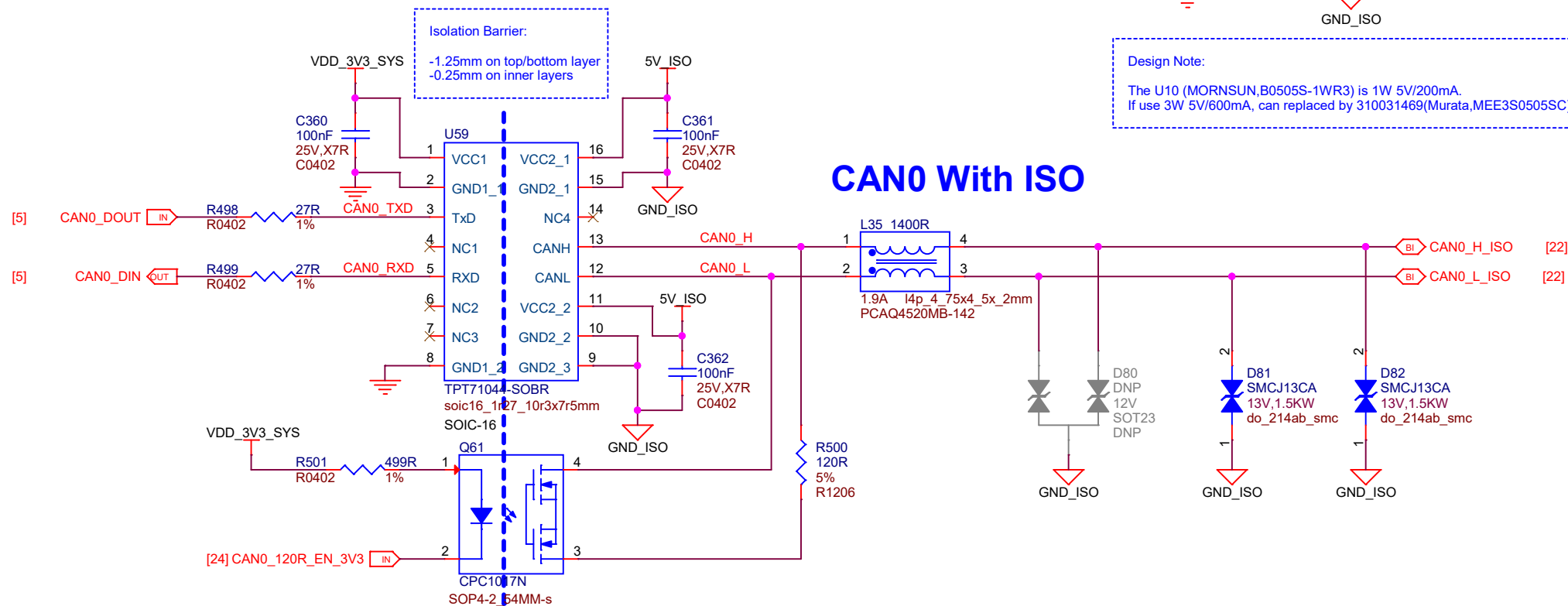
Digital Outputs (DO) rated 40V/0.2A



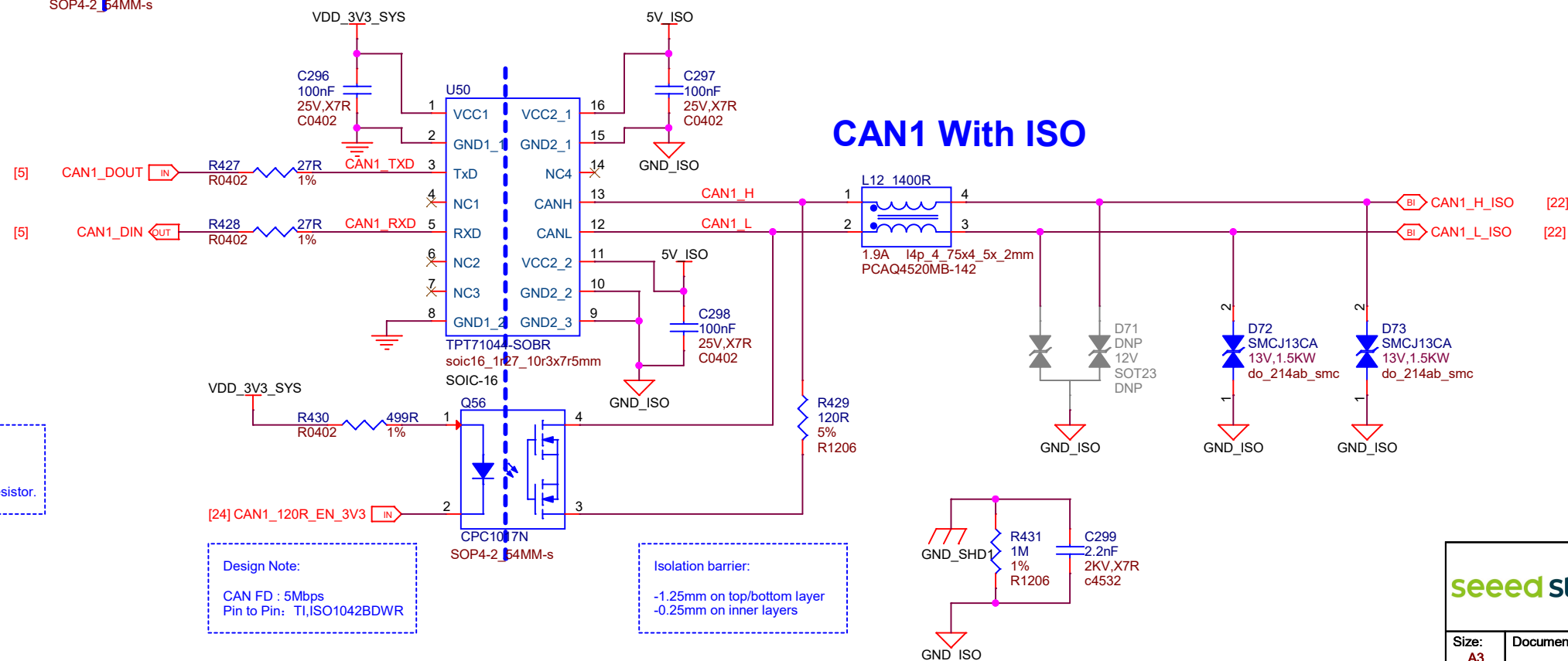
Isolated DCDC 5V to 5V

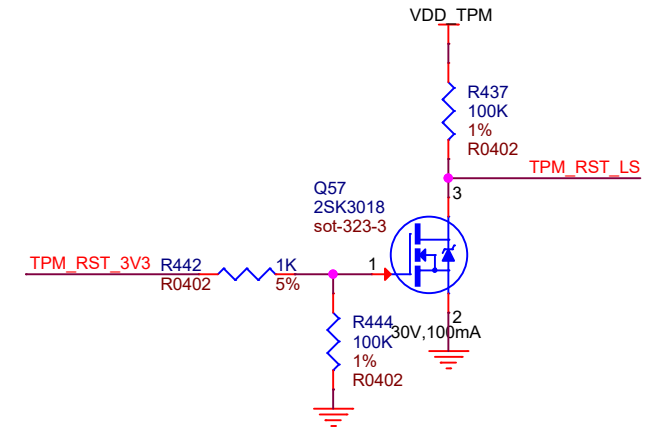
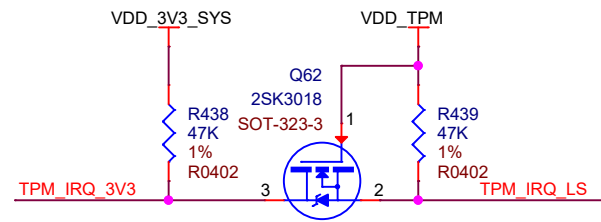
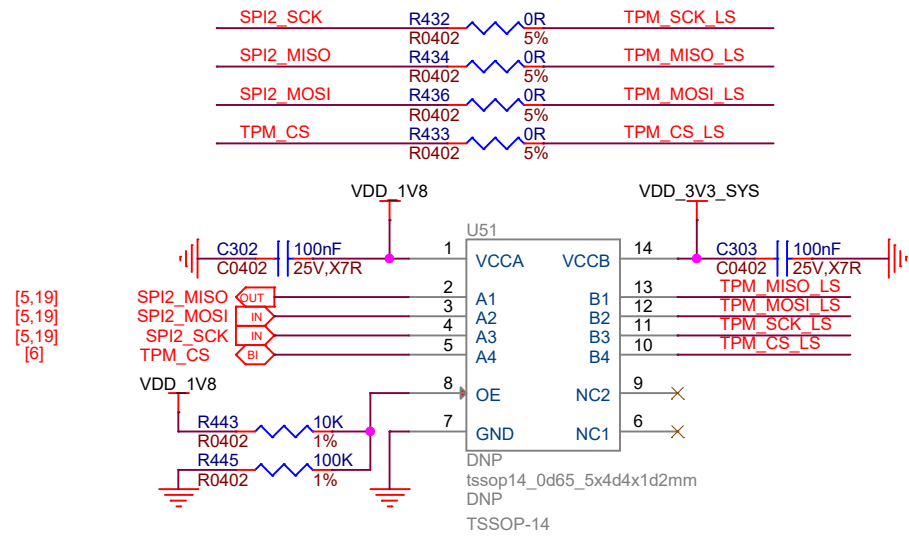


CAN0 With ISO

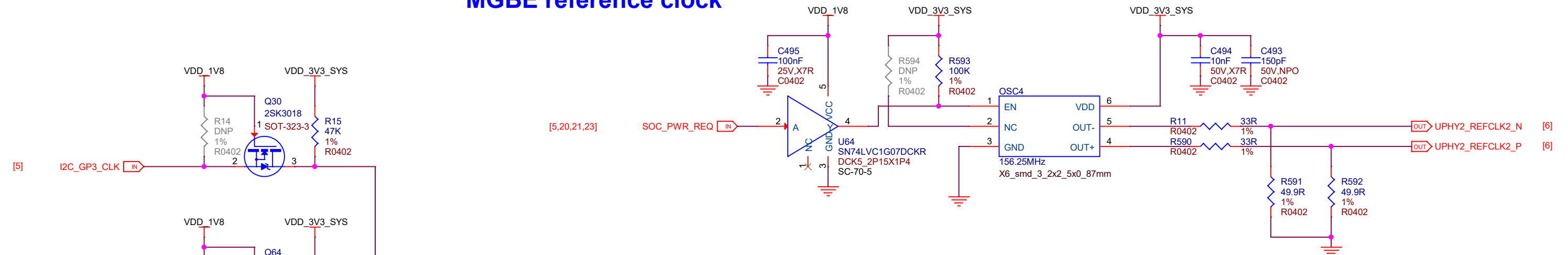


CAN1 With ISO

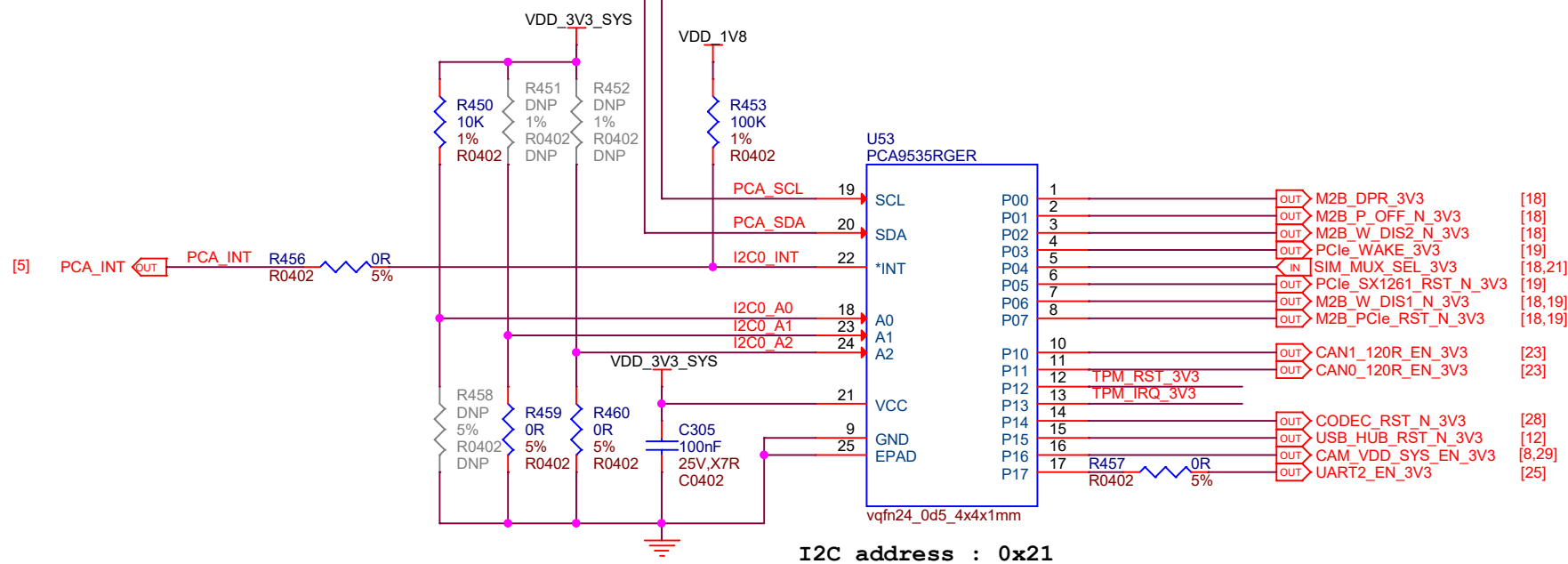




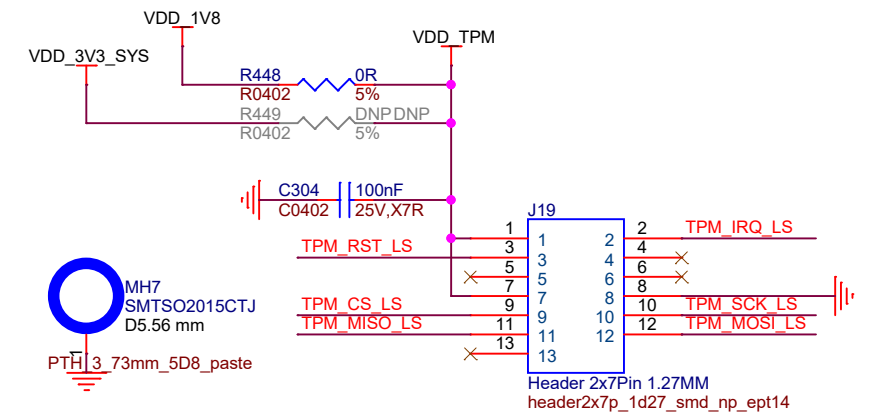
MGBE reference clock



IIC to GPIO



SPI TPM



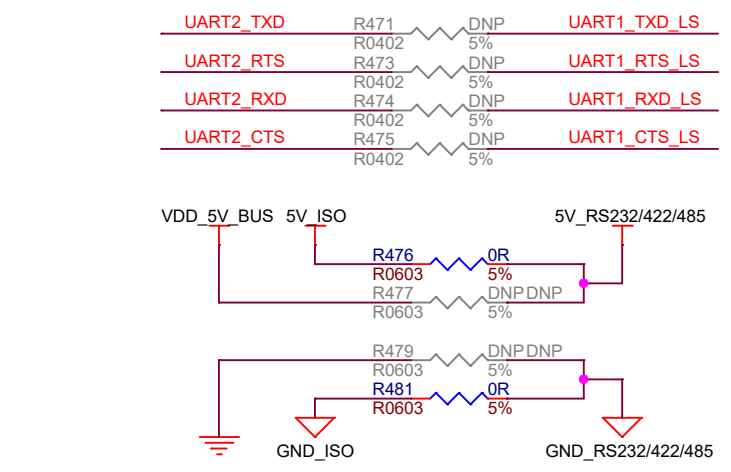
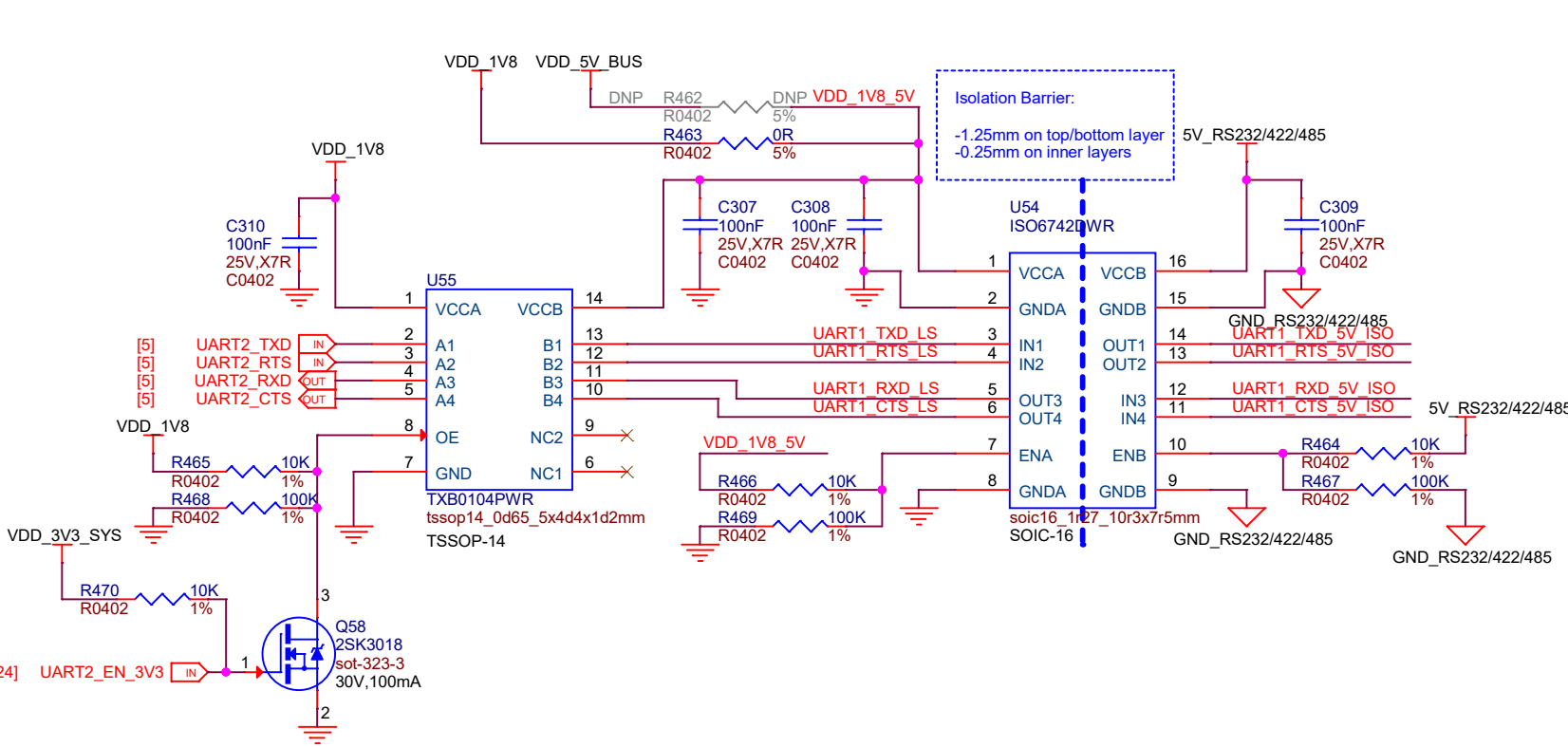
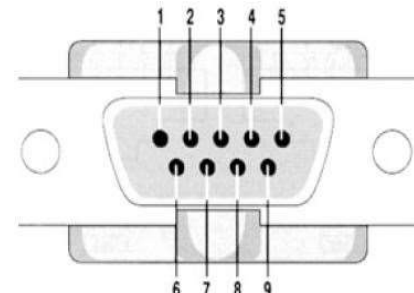
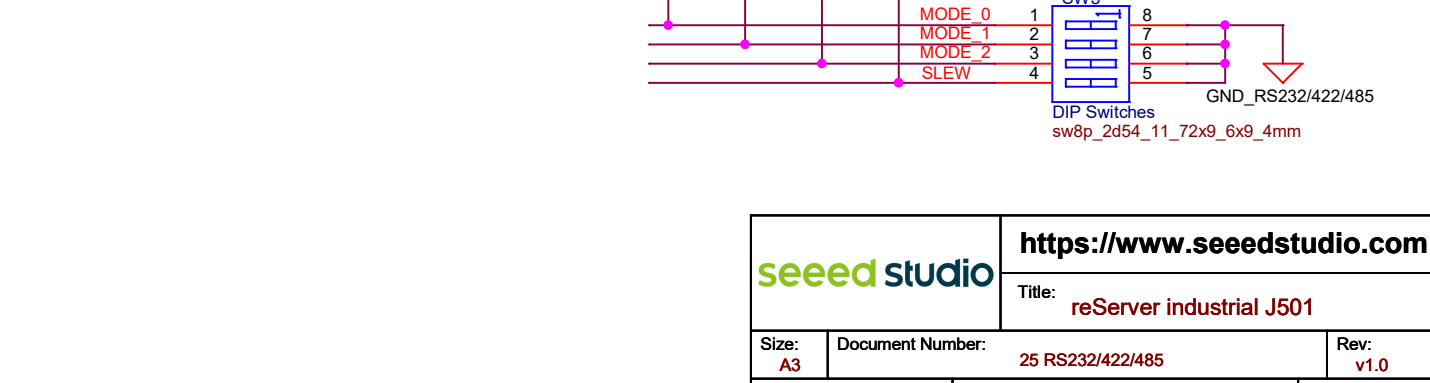
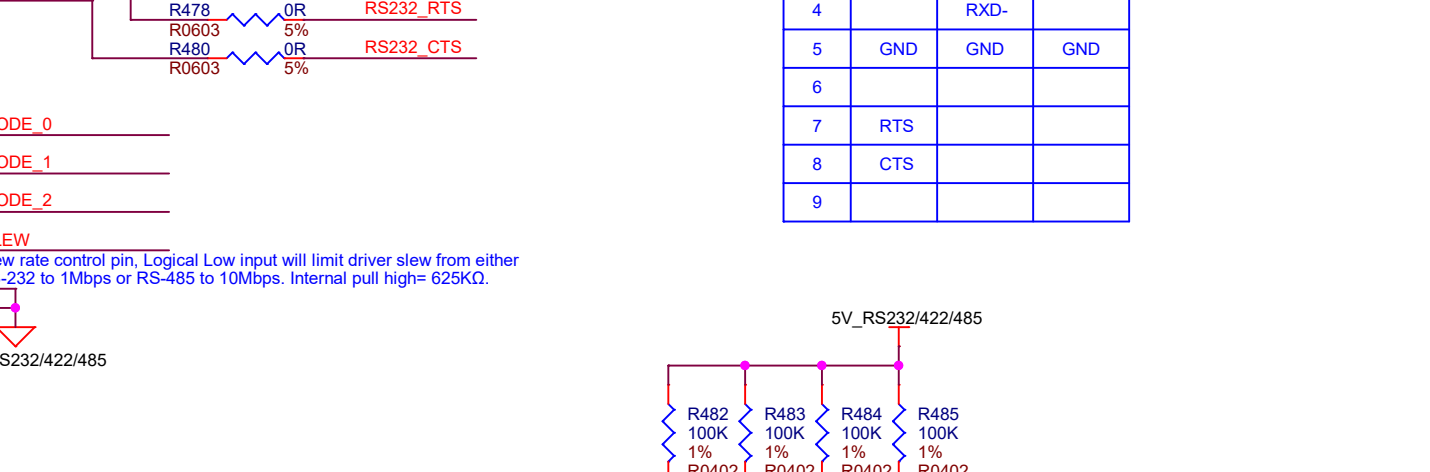
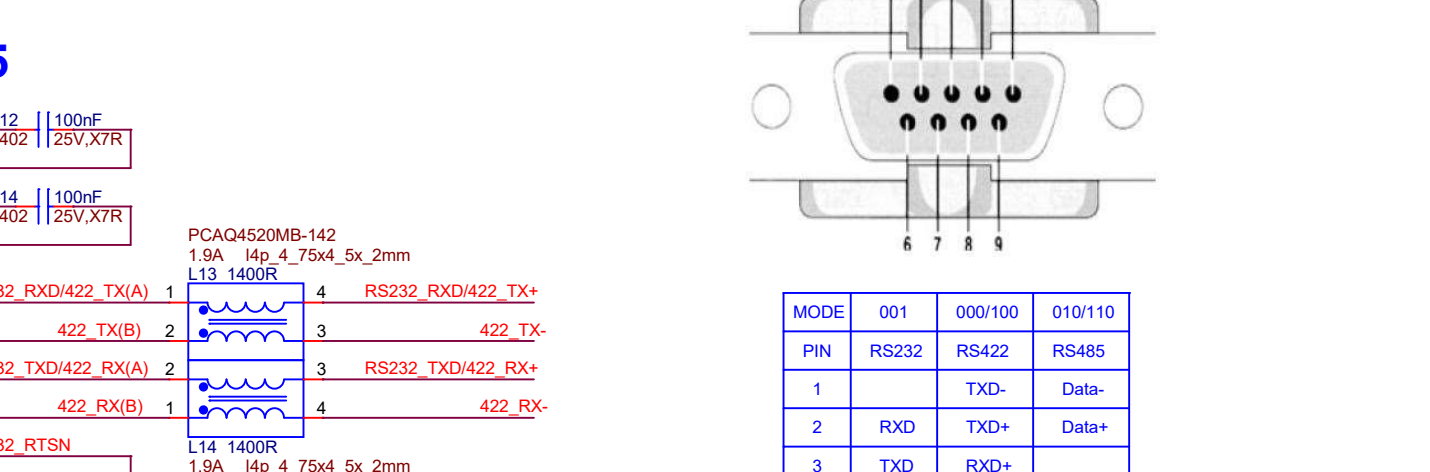
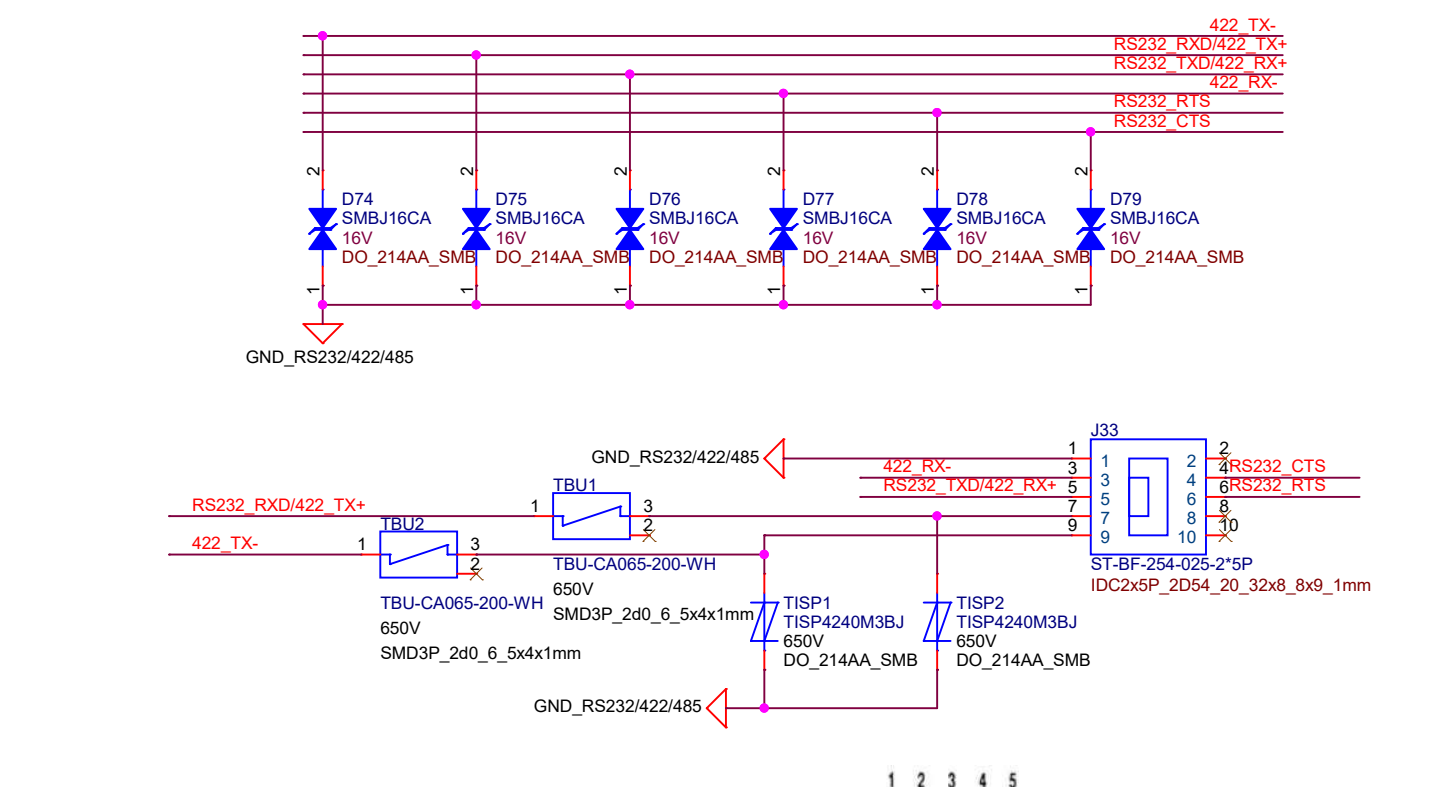
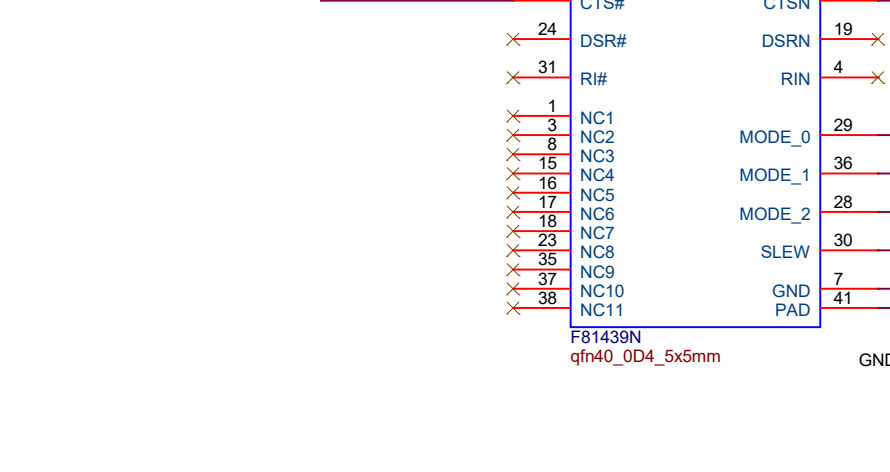
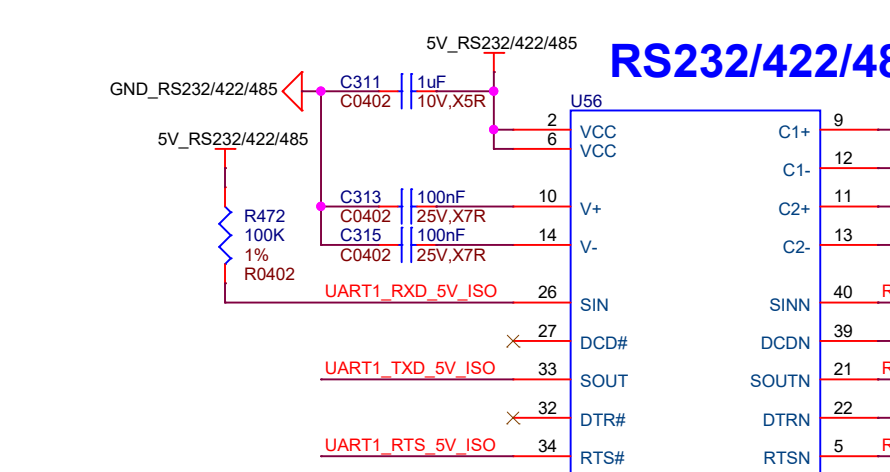
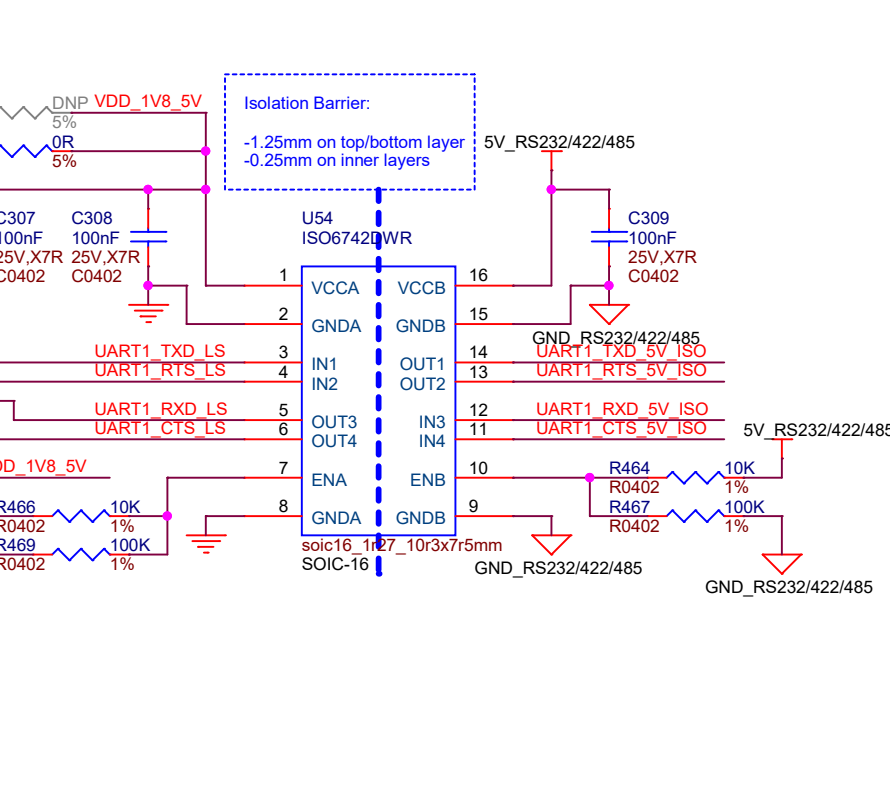
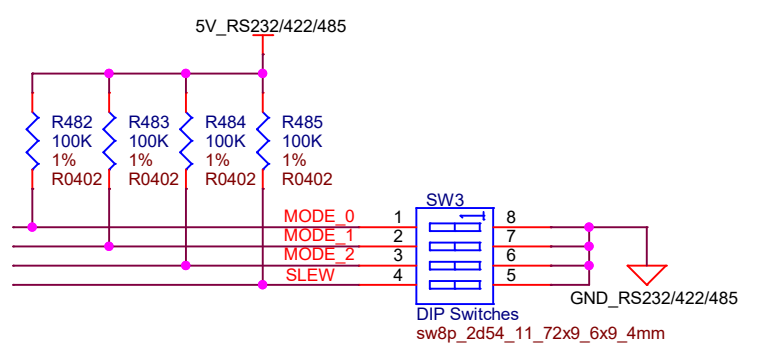


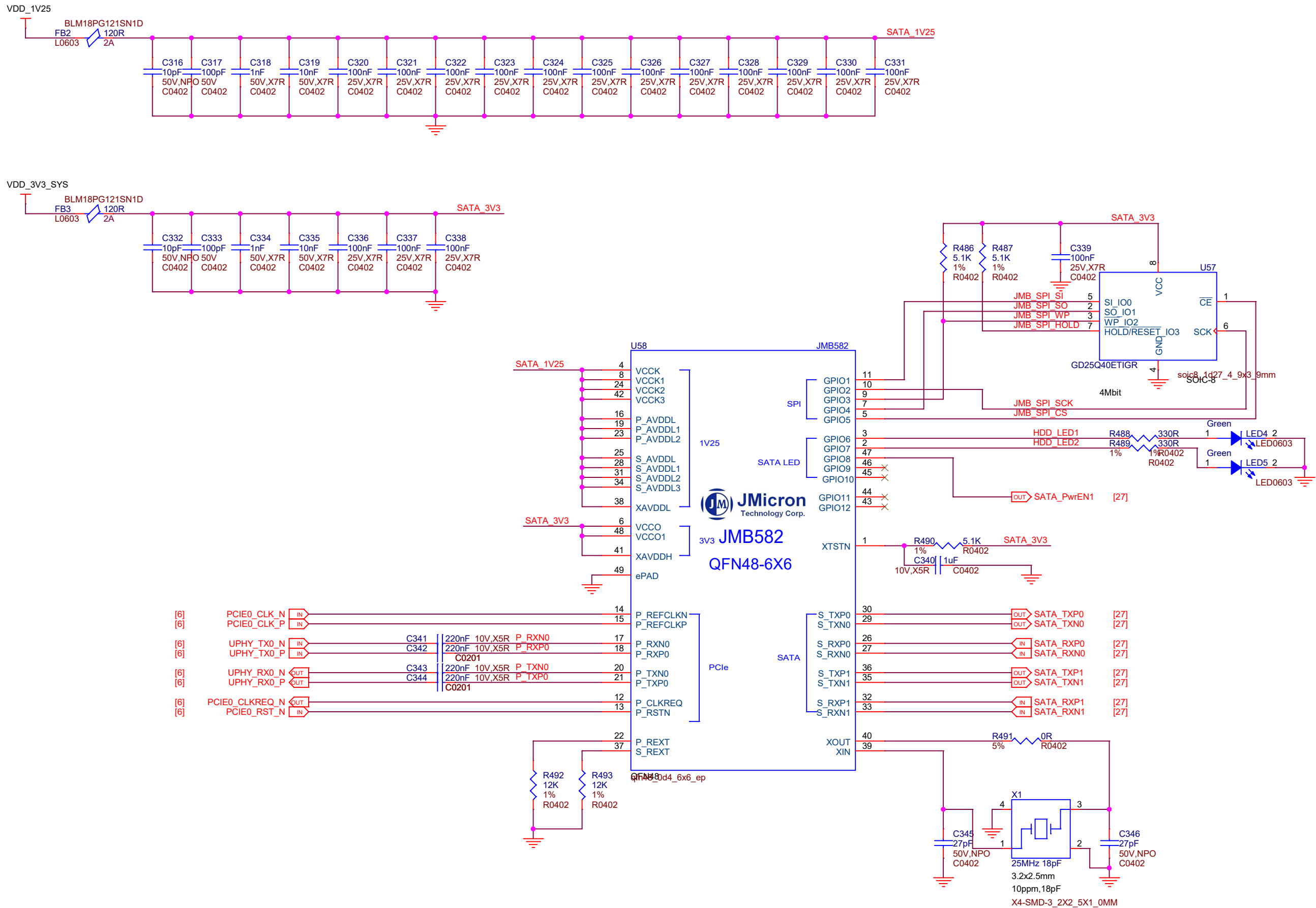
TABLE 1: Mode Select Configuration for F81439				
Pin 29 MODE_0	Pin 36 MODE_1	Pin 28 MODE_2	Mode	Status
0	0	0	RS-422 Full Duplex	1T/1R RS-422
0	0	1	Pure RS-232	3T/5R RS-232
0	1	0	RS-485 Half Duplex	1T/1R RS-485, TX ENABLE Low Active
0	1	1	RS-485 Half Duplex	1T/1R RS-485, TX ENABLE High Active
1	0	0	RS-422 Full Duplex	1T/1R RS-422 with termination resistor
1	0	1	Pure RS-232	1T/1R RS-232 co-exists with RS485 application without the need for the bus switch IC (for special usage).
1	1	0	RS-485 Half Duplex	1T/1R RS-485 with termination resistor TX ENABLE Low Active
1	1	1	Low Power Shutdown	All I/O pins are High Impedance



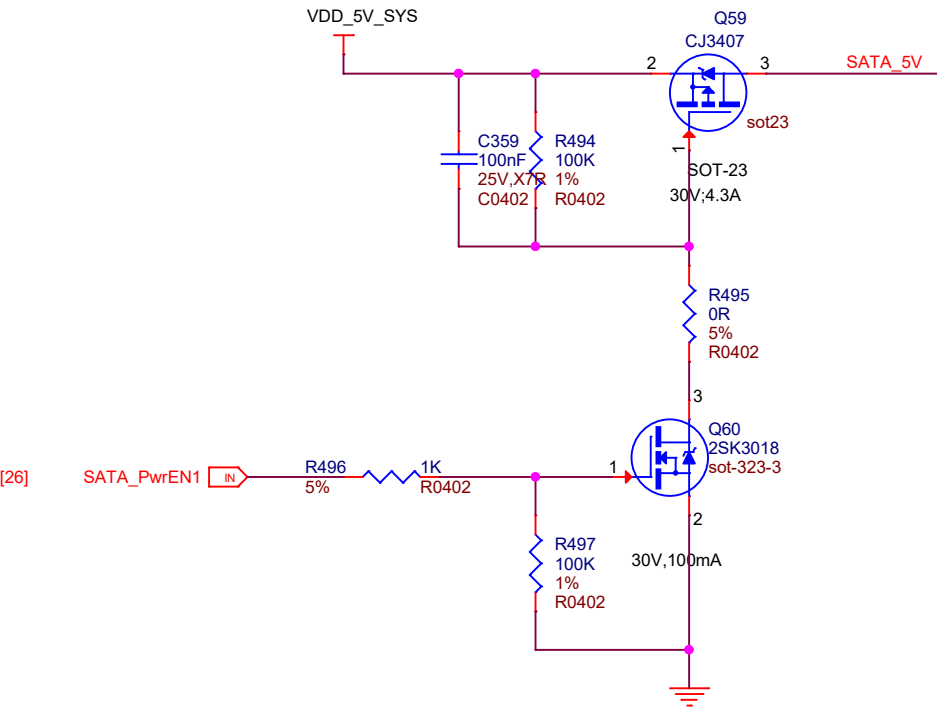
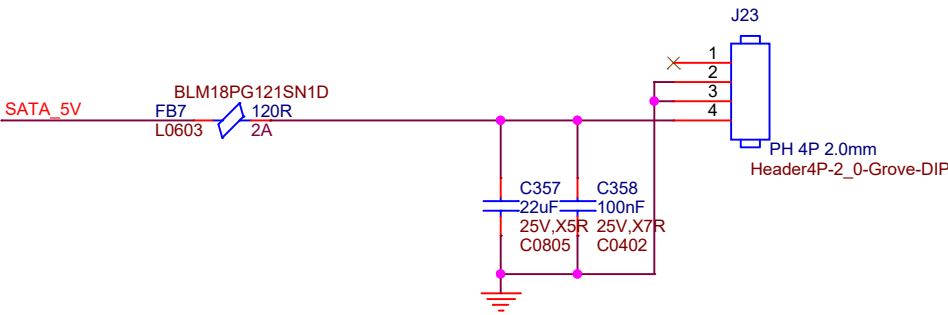
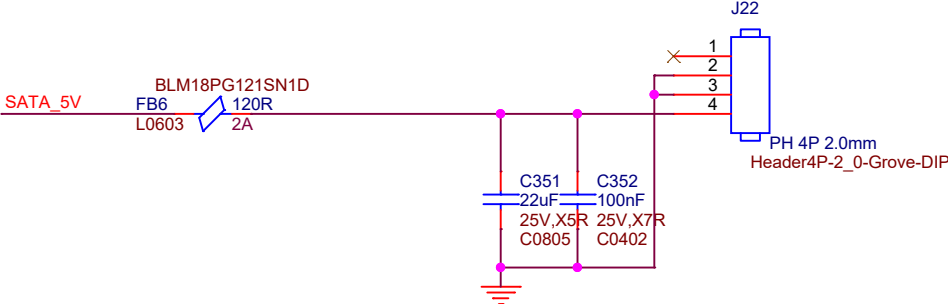
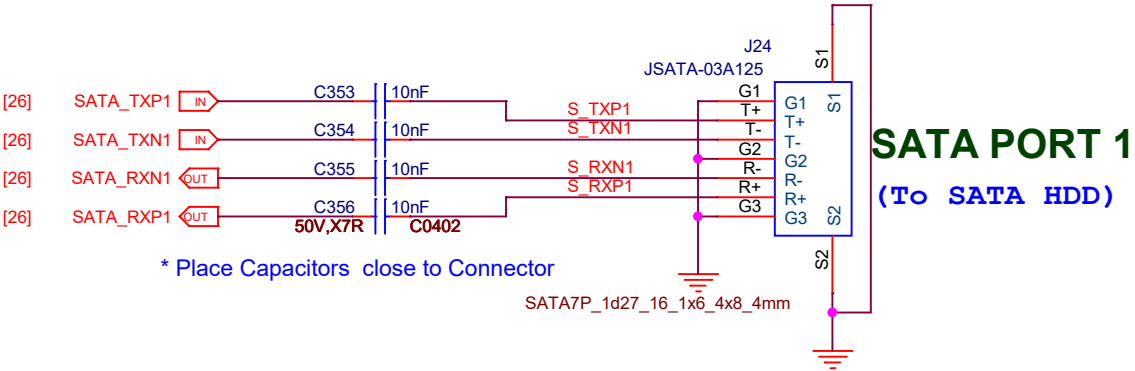
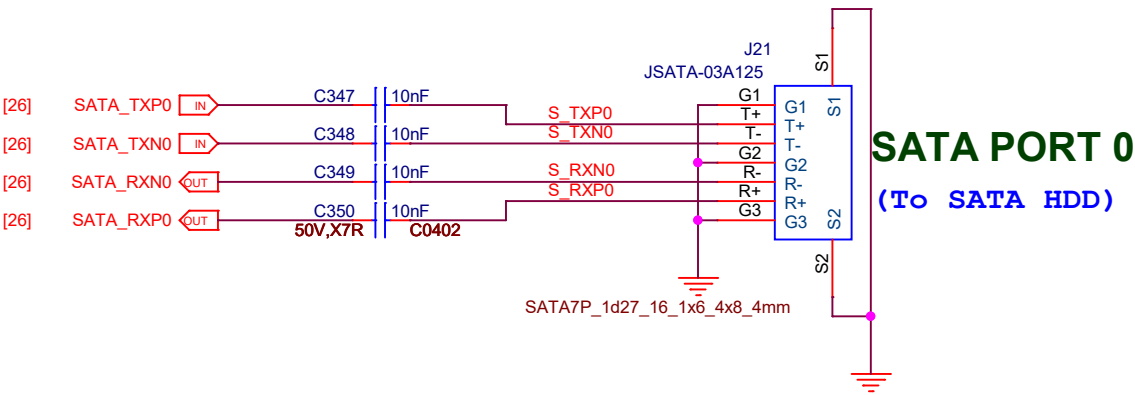
MODE	001	000/100	010/110
PIN	RS232	RS422	RS485
1		TXD-	Data-
2	RXD	TXD+	Data+
3	TXD	RXD+	
4		RXD-	
5	GND	GND	GND
6			
7	RTS		
8	CTS		
9			



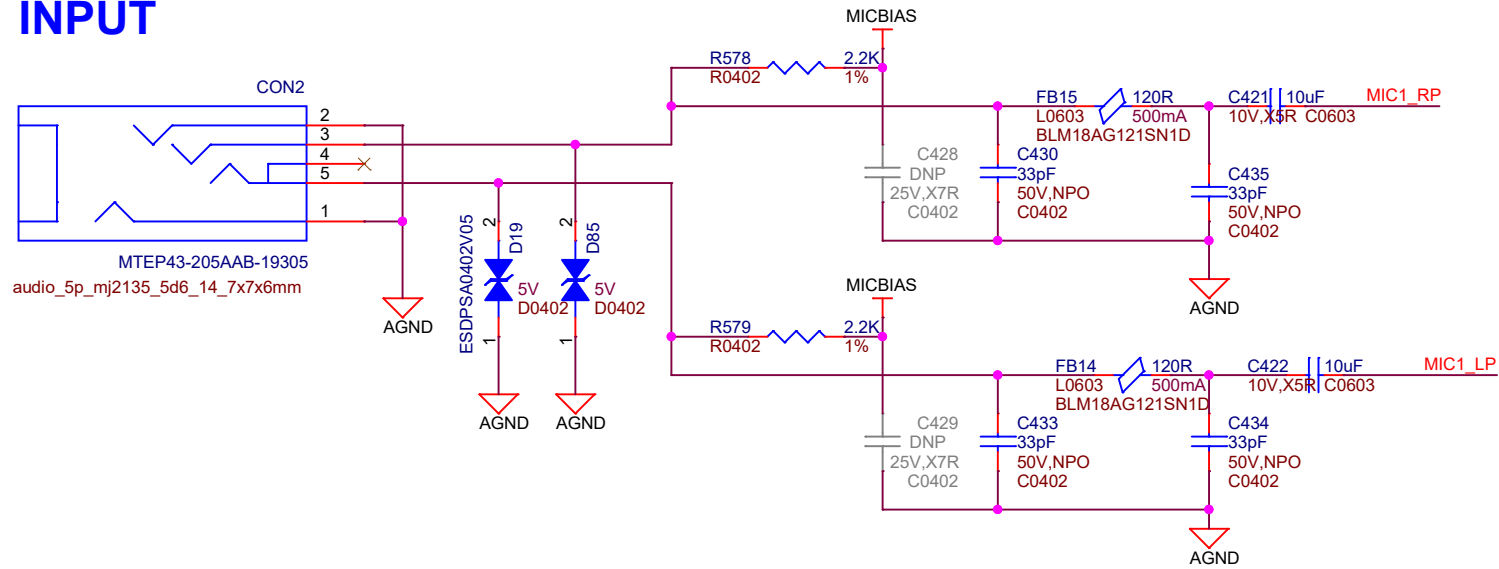
Sata Controller



2.5" Sata connector



INPUT



OUTPUT

